Visual Network tool: Individual belief networks from LLM-guided interviews and a visual canvas

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Abstract

Belief systems are theorized on individual and collective levels, but are typically studied and inferred only at the collective level. We introduce a new tool to measure individual-level belief networks from open-ended conversations, using a visual canvas. This "Visual Network" tool addresses several limitations of existing measures and enables fast and scalable measurement of the differences in content and structure of individual networks. Participants complete a semi-structured interview about their beliefs and behaviors on a given issue guided by an LLM, evaluate LLM-extracted statements that summarize the beliefs they have expressed, and construct personal belief networks by creating *supporting* and *conflicting* relations between these statements on a visual canvas. We validate the tool in a two-wave study on meat-eating. Participants evaluate the experience positively, and both belief network content and structure show satisfactory reliability across the two measurement points. We also find support for face, concurrent, and criterion validity of the measured individual belief networks. The new "Visual Network" tool opens up new empirical possibilities to test theories of belief dynamics and belief change.

Keywords: belief networks; belief dynamics; large language models; cognitive dissonance

1 Introduction

A growing number of theories of belief dynamics build on the concept of individual and collective belief networks (Bizyaeva, Franci, and Leonard 2022; Brandt and Sleegers 2021; Dalege, Borsboom, Van Harreveld, et al. 2016; Dalege, Galesic, and Olsson 2025; Goldberg and Stein 2018; Rodriguez, Bollen, and Ahn 2016; Schweighofer, Schweitzer, and Garcia 2020; Vlasceanu, Dyckovsky, and Coman 2024), whereby beliefs develop and change within a system of other beliefs about related issues, motivations, experiences, and behaviors. These belief systems can be represented as networks, allowing the use of ideas and methods from network science to formulate and test theoretical hypotheses about individual and collective beliefs. For example, this representation allows studying why some people's belief networks are more resilient (Scheffer et al. 2022), why some beliefs are more difficult to change than

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others (Brandt and Vallabha 2023; Turner-Zwinkels and Brandt 2022), how belief networks change to adapt to internal and external pressures (Dalege, Borsboom, Van Harreveld, et al. 2016; Dalege and Does 2022; Turner-Zwinkels and Brandt 2022), and how network structure relates to decision and action (Brandt, Sibley, and Osborne 2019; Dalege, Borsboom, Harreveld, et al. 2017; Hasan et al. 2025). However, theorizing about belief systems as networks has been impeded by the difficulty of measuring belief networks at the individual level.

Most existing studies rely on belief networks inferred from population-level data, where relationships between beliefs are obtained from correlations between individuals (Boutyline and Vaisey 2017; Dalege, Galesic, and Olsson 2025; Turner-Zwinkels, Johnson, et al. 2021). While valuable for understanding how beliefs affect each other and predict behavior (Chambon, Dalege, Borsboom, et al. 2023; Dalege, Borsboom, Harreveld, et al. 2017) the population-level approach has important conceptual limitations. Most theories of belief networks operate on the level of individual beliefs, and population-level estimates of belief networks might not represent individual belief networks well (Brandt 2022; Brandt and Morgan 2022). Furthermore, by estimating one shared (collective) belief network they obscure structural variation at the individual level. Most fundamentally, some individuals might have more, and others fewer, beliefs about a given issue. For example, the topic of meat-eating might prompt some people to think mostly about taste (Piazza et al. 2015), while others might relate the topic to a wider set of beliefs, for example, about health effects and animal welfare concerns (Hopwood et al. 2020). In addition, individual belief networks can be more or less interconnected. For example, people with more political knowledge might have more interconnected beliefs about political issues than those with less knowledge (Brandt 2022). Individual differences in size, connectivity, and other aspects of network structure could constrain the way people adjust their beliefs in light of new evidence and changes of related beliefs (although see Brandt and Vallabha (2023)).

Measuring individual belief networks has been difficult because it requires either multiple measurements of the same beliefs of an individual or pairwise comparisons of different beliefs of an individual. The first method–longitudinal measurement–asks about the same beliefs repeatedly over several time points, and infers relationships between these beliefs from the patterns of their co-change over time (Chambon, Dalege, Waldorp, et al. 2022; Chambon, Dalege, Borsboom, et al. 2023). However, estimating individual-level belief networks from longitudinal data relies on individual-level belief change, and is challenging because beliefs tend to be relatively stable over time (Kiley and Vaisey 2020), and estimating temporal networks might require upward of 100 observations even for a few variables (Mansueto et al. 2023). In addition, longitudinal measurement of the same individuals can change their belief systems just because they keep on responding to the same questions over and over again (Bartels 1999). The second method–pairwise measurement–asks individuals to make conceptual similarity judgments about beliefs to measure the individual-level network structure (Brandt 2022). However, this method can become cognitively and financially costly when researchers are interested in relationships of more than a few beliefs, since participants are asked about each pair of beliefs separately. In addition, the mere act of asking about a connection between two beliefs might induce such a relationship even if a participant did not think about it before (Tourangeau, Rips, and Rasinski 2000). Finally, pairwise comparisons by definition allow only for evaluating each relationship in isolation of the whole belief system.

More generally, most previous methods to obtain belief networks still rely on pre-specified questionnaires. For example, beliefs about genetically modified food (GM) might be measured using a questionnaire designed to reflect the theoretically assumed moral and social beliefs relevant to GM food (Dalege and Does 2022; Dalege, Galesic, and Olsson 2025). While enabling easy inter-individual comparisons, such pre-specified questionnaires assume or impose the same number of beliefs on all participants, even when some participants might have never considered a particular belief before. The mere act of asking about a belief can change a person's belief system, as individuals may form a belief about an issue on the spot when asked (Schwarz 1999; Tourangeau, Rips, and Rasinski 2000). Further, some individuals might have many more beliefs, and more nuanced beliefs than a pre-specified questionnaire has envisioned. Individual belief networks measured in this way can appear to be more similar than they really are and as a result be less predictive of belief change and behavior than they could be.

Here we introduce a new measurement methodology, the "Visual Networks" (VN) tool, to elicit individual belief networks through a combination of LLM-assisted semi-structured interviews and a visual canvas on which participants indicate how they perceive beliefs to relate. The VN tool mitigates some of the conceptual and practical shortcomings of previous measures. It allows individual belief networks to be measured and to vary in content, size, and structure. In addition, it avoids some of the measurement and scalability issues of existing individual-level methods. Specifically, it reduces the cognitive and financial costs of measuring individual belief networks by a visual canvas on which individuals can easily and quickly report the relationships between their beliefs. And, rather than relying on pre-specified questionnaire items, the VN tool capitalizes on recent advances in large language models (LLMs) to elicit beliefs in an open-ended manner and non-suggestive manner, allowing participants to report the beliefs they consider relevant in their own words. This opens doors to large-scale empirical measurement of individual belief systems and further theory development and testing. In what follows, we describe the new tool and show that it has satisfactory psychometric properties.

2 Description of the Visual Network Tool

The VN tool consists of three main phases (see Figure 1). The first phase is a semi-structured interview with an LLM interviewer. The second phase involves LLM-extraction of short summary statements (network nodes) from this interview transcript. Participants rate these statements based on their accuracy and relevance. In the third phase, participants create their personal belief networks on a two-dimensional visual canvas using these summary statements. In addition to these three main phases, we also develop a training procedure that can be used to teach participants, if needed, how to draw connections on the canvas. For validation purposes in this study, we also include sections in which participants evaluate pairwise relationships for a subset of belief pairs, assess how accurately various network representations describe their beliefs, and answer several more standardized questions about their beliefs.

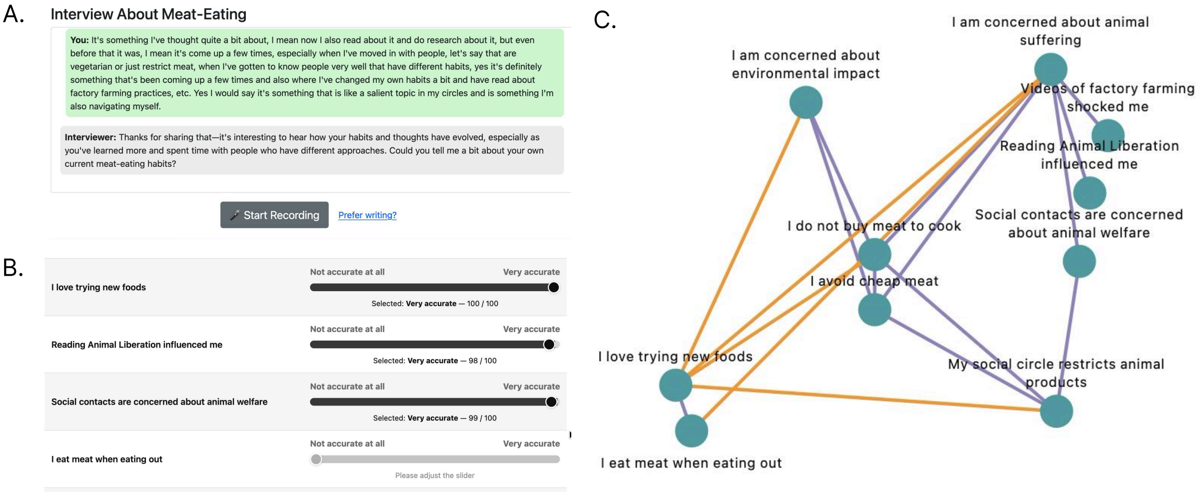


Figure 1: Overview of the tree main phases of the VN tool. All screenshots from lead author completing the survey on 2026-03-04. **A.** screenshot of the semi-structured LLM interview about meat-eating. In the default mode, participants record their answers. These are live-transcribed with `whisper` but can be edited by the user before they are sent. Alternatively, participants can also choose to write their answers instead. **B.** participants rate LLM-extracted summary statements based on *accuracy* and *relevance*. **C.** participants create *supporting* (purple) and *conflicting* (orange) connections between summary statements using the canvas tool.

In this initial study, we apply the VN tool to the topic of meat-eating. This is a suitable domain for our purposes because many people have somewhat conflicting beliefs about meat-eating. For example, while many people enjoy the taste of meat, some are concerned about the effect on personal health and the environment (Beardsworth and Bryman 2004), resulting in "meat-related cognitive dissonance" (Rothgerber and Rosenfeld 2021). Beliefs and behaviors around meat-eating are likely to be relatively stable over time, enabling us to assess reliability of our measures over a one-week interval.

2.1 LLM semi-structured interview

The first part of the survey instrument consists of a semi-structured LLM interview. Our guiding design principle is that the LLM should never be suggestive or ask about specific beliefs (e.g., animal welfare). Instead, the LLM is broadly instructed to explore what is salient to the meat-eating habits of the participant, with the objective to cover at least (1) the meat-eating habits of the interviewee, (2) the personal motivations that the interviewee has to eat meat or avoid eating meat, (3) the meat-eating habits of the social contacts of the interviewee, and (4) the motivations social contacts have to eat meat or avoid eating meat. Of course, any other domain could be explored instead of meat-eating. Our setup here results in interviews with some shared semantic content, but also allows interviews to focus on what is particularly important to individual participants.

Participants can either speak (with live-transcription using `whisper`) or they can type out their answers. An LLM (`gpt-4.1-2025-04-14`) is given 8 turns (questions) to learn about the meat-eating beliefs of the participant. The length or duration of the interview can be adjusted as the researcher sees fit. After each participant answer, the LLM receives an updated transcript and generates the next interview question. The interview protocol and design is inspired by recent work in conducting semi-structured interviews with LLMs (Park et al. 2024; OpenResearch 2025a). The interface is shown and described in 7.1.1, the choice of LLM is discussed in 7.2, and the full LLM-prompt for the interview is shown in SI Section 7.4.1.

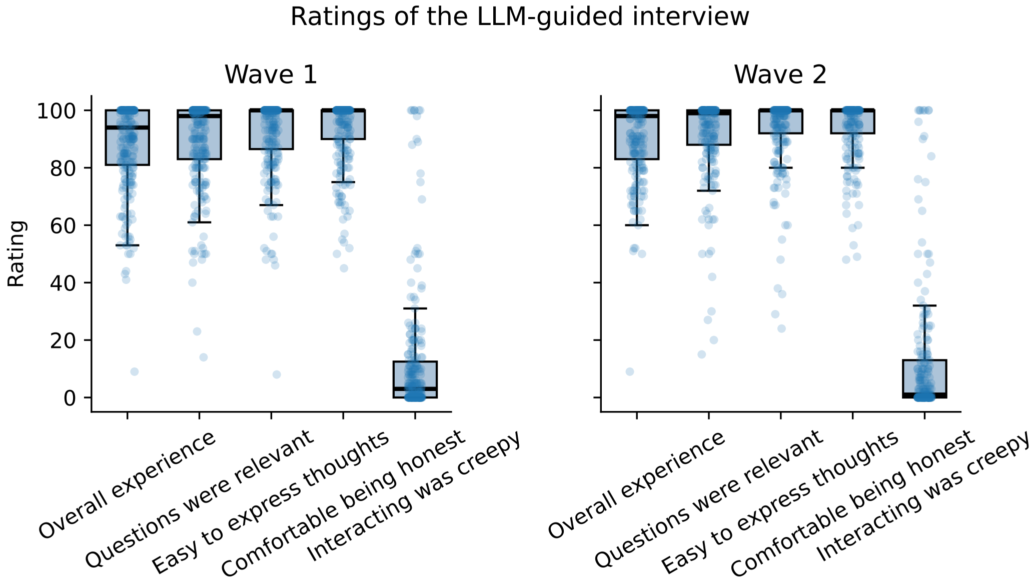


Figure 2: Participant ratings of the semi-structured LLM interview for wave 1 ($n = 247$) and wave 2 ($n = 217$). The complete questions and scale is described in SI Section 7.1.1.

2.2 Summary statements

The second phase of the VN tool starts with an LLM-extraction of summary statements from a transcript of the interview described in the previous section. This step condenses the interviews into a small set of statements that we can treat as network nodes. LLMs are particularly well suited for text summarization, which has previously been exploited in the domain of beliefs and attitudes (Velez, P. Liu, and Clifford 2025). Here, we aim to extract an open set of concise and short descriptions (summary statements) of motivations and behaviors that came up in the interview–summary statements that are both "accurate" reflections of what was said, but also "relevant" to the meat-eating habits of the interviewee. To achieve this, the complete transcript of questions and answers from the interview is provided as the input to an LLM (`gpt-4.1-2025-04-14`). The model is instructed to return a maximum of 10 summary statements. We impose this constraint because of the downstream canvas task (described below), but this number is otherwise arbitrary and can be dropped depending on downstream goals. The prompting strategy was developed based on early pilot testing. The LLM-prompt for this step is documented in SI Section 7.4.2.

After LLM-extraction of summary statements, participants evaluate their personally extracted summary statements on two dimensions; *accuracy* and *relevance*, both measured on a slider-bar scale from 1 ("Not at all") to 100 ("Very"). For the canvas task (described below) only summary statements rated by participants as at least "Accurate" (60 on the 100-point scale) are eligible. In addition to the (maximum) 10 summary statements extracted by the LLM, we expose participants to two fake summary statements. This is done to test participants' attention and reasoning. If participants rate the odd one of the two ("My friends often go out to eat seahorse") as "somewhat accurate" or above (at least 40 on the 100-point scale) in any of the two waves of the survey, we filter them out for inattention. The interface design for this part of the survey instrument is described in detail in SI Section 7.1.2.

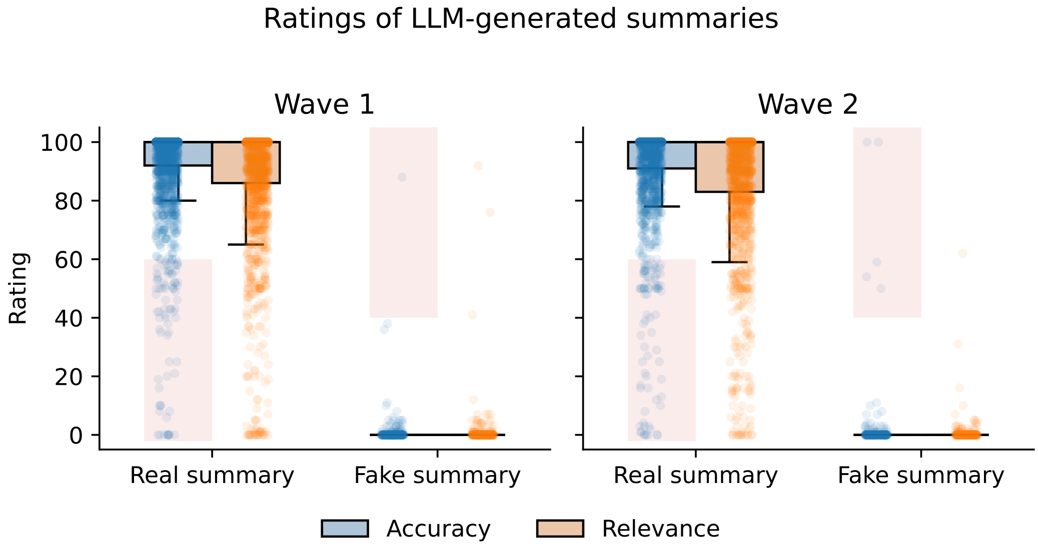


Figure 3: Participant ratings of the summary statements extracted by the LLM from participant interviews and the attention check that we use to filter out inattentive participants. Figures are based on $n = 247$ participants for wave 1 and $n = 217$ participants for wave 2. Statements rated as less than “Accurate” (60 or below on the 100-point scale) are filtered out (shown in red). Participants that rate odd one of the attention checks (fake summary) as at least “Somewhat accurate” (40 or above on the 100-point scale) are excluded from further analysis (shown in red).

2.3 Visual canvas task

The last part of the VN tool is for participants to code the relations between their personal beliefs using a visual canvas task. This consists of three parts (screens) that participants complete. All three parts are documented with screenshots in SI Section 7.1.3 (Figures 13, 14, and 15). First, participants arrange their summary statements such that beliefs that “go together” are closer to each other, and beliefs that do “not go together” are further apart on the canvas. This corresponds closely to classic spatial similarity tasks in cognitive science (Hout, Goldinger, and Ferguson 2013). Participants drag the summary statements that were extracted specifically based on their interview into the canvas from the side of the screen (see Figure 13). This first part of the canvas task serves two purposes; (1) participants have time to mentally organize their statements and consider how they relate, (2) the spatial organization may be analyzed as a proxy for how closely items are related.

In the second part of the canvas task, participants create connections between summary statements that “support each other” (Figure 14) and in the third part of the canvas task, they independently create connections between summary statements that “conflict with each other” (Figure 15). We chose “support” and “conflict” because we wanted some indication of (in)consistency between beliefs while introducing as few assumptions as possible as to where this inconsistency stems from. Different beliefs might be connected logically or culturally (Brandt and Vallabha 2023), but many possible types and constraints could be possible (also e.g., causal or semantic). Here we wish to include whichever type of connection can be broadly understood as either "conflicting" or "supporting". To create a connection, participants click on one summary statement node, which highlights the node, and then by clicking on another summary statement node, which completes the connection (and the same to remove a connection).

2.4 Training

Before participants do the canvas task (described above), they go through training examples. We design 3 training examples to allow participants to get familiar with creating "conflicting" and "supporting" connections between beliefs (a notion which might not be familiar to participants) and teach them how to use the canvas to code such relations. Participants complete all three training examples in the first wave, and complete one randomly drawn training example in the second wave. Training is structured as follows; Participants first read a short vignette describing a person with some goal (e.g., learning a language) along with motivations and challenges. Then they (1) spatially arrange the provided summary statements, (2) create supporting connections, and (3) create conflicting connections. As such, the training task exactly mirrors the downstream canvas task, just with made up vignettes rather than the participant's own personal meat-eating statements. We design the vignettes and the summary statements in such a way that some statement pairs should be coded as "supporting" and "conflicting", which allows us to provide live feedback to participants such that they can learn how to both operate the tool, and learn how to think about connections. Training materials and screenshots are provided in SI Section 7.1.4.

2.5 Network Comparison

To evaluate whether participants visually recognize their own network (as accurate) we design a network comparison task. Here, we ask participants to evaluate and compare four networks. First, the network they created using the visual canvas task. Second, a network extracted during the survey with an LLM (`gpt-4.1-2025-04-14`) based on participant interview transcripts and the same summary statements that participants use to create their own belief network. The third and fourth networks are the same as the two first ones, but with randomly perturbed connections between summary statements. Participants are shown two of these four networks at a time and asked to rate how well each network describes their meat-eating beliefs. More details and screenshots are provided in SI Section 7.1.5. Participants only complete this task during the second wave of the study.

2.6 Pairwise Interview

To assess concurrent (or construct) validity, we additionally measure "supporting" and "conflicting" connections for a subset of beliefs by pairwise comparison at the end of wave 2. We sample up to nine pairs of statements, aiming to include three pairs that were coded as supporting, three pairs that were coded as conflicting, and three pairs that were not coded by the participant in the canvas task. In an LLM-assisted interview (`gpt-4.1-2025-04-14`), participants are first asked an open question about the two statements (e.g., "In your own words, how does x and y relate to each other, if at all?"), and then a closed question ("Thank you for explaining that. If you had to choose, how would you describe the way these two things relate overall?). For this second question, participants are given three options that map to categories of “support”, “conflict”, and “no connection”. More details and screenshots are provided in SI Section 7.1.6.

3 Validation study

Measuring individual belief networks raises specific psychometric challenges. Classical psychometric instruments typically define reliability and validity for sets of items that are assumed to be interchangeable indicators of a latent construct (Cook, Campbell, and Shadish 2002). In contrast, the network approach adopted here treats beliefs as distinct entities and takes the pattern of relations among them as the primary object of analysis. Because the present tool relies on open ended responses to elicit beliefs, the same or similar beliefs may be expressed using different words across measurement occasions. As a result, standard criteria such as internal consistency or agreement among items are not informative for evaluating this type of data. Instead, psychometric evaluation needs to address whether the inferred network structure itself is reliable, interpretable, and predictive.

We evaluate psychometric properties of the VN tool in a two-wave study of the same participants. The interval between the two waves is only one week in order to avoid confounds due to actual changes in belief systems (DeVellis and Thorpe 2021). We focus on topic of meat-eating, where we expect beliefs to be relatively stable, at least over short time periods (Beardsworth and Bryman 2004). Participants were recruited from the crowd-sourcing platform Prolific. The first wave (started on December 9, 2025) was completed by 247 participants, of which 217 participants completed the second wave (started on December 16, 2025). After excluding participants who failed the most obvious attention check (asking participants whether it is accurate that their friends often go out to eat seahorse) or had incomplete records, we were left with 210 participants included in the analyses reported next.

In the sections below, we assess (i) the reliability of interview engagement and basic network properties, and (ii) the reliability of semantic content of elicited beliefs across two waves, taking into account that beliefs are reported in participants’ own words. We further assess (iii) face validity of the measured belief networks based on participants’ judgments of whether the elicited beliefs and their relations accurately reflect their belief networks, (iv) concurrent validity of network relationships by comparing those obtained with our tool to those measured using the pairwise comparison method, and (v) criterion validity, by testing whether network centrality measured in the first wave predicts persistence of beliefs in the second wave.

4 Results

4.1 Reliability of interviews and network properties

We first compare the interviews and basic network properties observed on two measurement occasions. First, we observe a strong correlation ($r = 0.75$) between the total number of words spoken or typed at wave 1 and wave 2 within individuals. Comparing the number of summary statements (network nodes) extracted by the LLM from the interviews shows a moderate but reliable correlation ($r = 0.38$). During the survey, we artificially limit the number of summary statements extracted to a maximum of 10, here we show the number extracted using the same interviews and LLM-prompt but without this constraint. Finally, we assess the relationship between the number of connections (network edges) that participants create between nodes on the canvas in each wave, and find a moderate correlation ($r = 0.54$). This result is obtained from the set of summary statements that is extracted during the experiment, which is limited to 10 (see Section 2.2). Detailed results for all three comparisons across waves are shown in Figure 4.

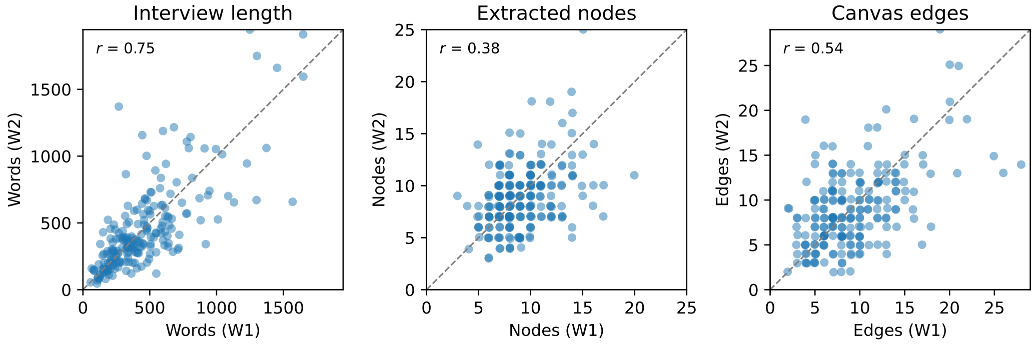


Figure 4: *Left:* Total number of words spoken by, or typed, by participants in wave 1 (mean 456.1 words) and in wave 2 (mean 450.0 words). Distributed over 8 questions this gives an average of 55-60 words per answer. *Middle:* Total number of summary statements (network nodes) extracted by the LLM for each participant in wave 1 (mean: 9.2) and wave 2 (mean: 8.9). Result obtained without the maximum constraint of ten nodes. *Right:* Total number of connections (network edges) created by each participant on the canvas in wave 1 (mean: 9.4) and wave 2 (mean: 9.1).

4.2 Reliability of semantic content

To compare the semantic content of belief networks across measurement occasions, we use BERTopic to map summary statements to a common set of topics (Grootendorst 2022). BERTopic combines (i) transformer-based embeddings, (ii) dimensionality reduction (UMAP), and (iii) density-based clustering (HDBSCAN). To avoid biases in conclusions due to specifics of the topic model, we compare results for 10 different topic model fits, using different sentence-embedding models and hyper-parameters (see SI Section 7.3 for a detailed description and SI Table 1 for model comparison). We focus on the results for the `qwen3-embedding-0.6b` sentence-embedding model run with hyper-parameters `n_neighbors`= 30, `n_components`=2, and `min_cluster_size`= 50. This model was chosen because, in comparison with other BERTopic fits it has a reasonable number of topics, and relatively few topics assigned as outliers. This BERTopic scores best on well-established quality metrics of density-based clustering (Moulavi et al. 2014), and it was manually evaluated to separate expected topics well (see SI Section 7.3). Our approach and selection criteria are inspired by Ha et al. (2025).

To assess the semantic overlap between any two sets of nodes, both within participants (same participant at wave 1 and wave 2) and across participants (one participant at wave 1 and another participant at wave 2), we compute the Phi coefficient (Matthews correlation for binary data) which is given by:

$$\phi = \frac{ad - bc}{\sqrt{(a+b)(a+c)(b+d)(c+d)}}$$

where letters denote counts of times a topic was present in both (a), only in the first (b), only in the second (c) or in neither (d) of the two waves. We always compare across the two waves, both for within-participant calculations and for across-participant calculations. Within individuals (self–self pairs across the two waves) we have $\bar{\phi}_{self} = 0.36$ and across individuals (self–other pairs) we have $\bar{\phi}_{other} = 0.10$ for the selected model. Across all 10 topic model fits, we observe $\bar{\phi}_{self}$ between 0.34 and 0.43 (see SI Table 1). Standard categories classify ϕ coefficients $\in \{.30, .39\}$ as moderate and $\in \{.4, .49\}$ as strong. We show the distribution of ϕ scores across individuals (self–self: $\bar{\phi}_{self}$, and self–other pairs: $\bar{\phi}_{other}$) for the selected model in Figure 5.

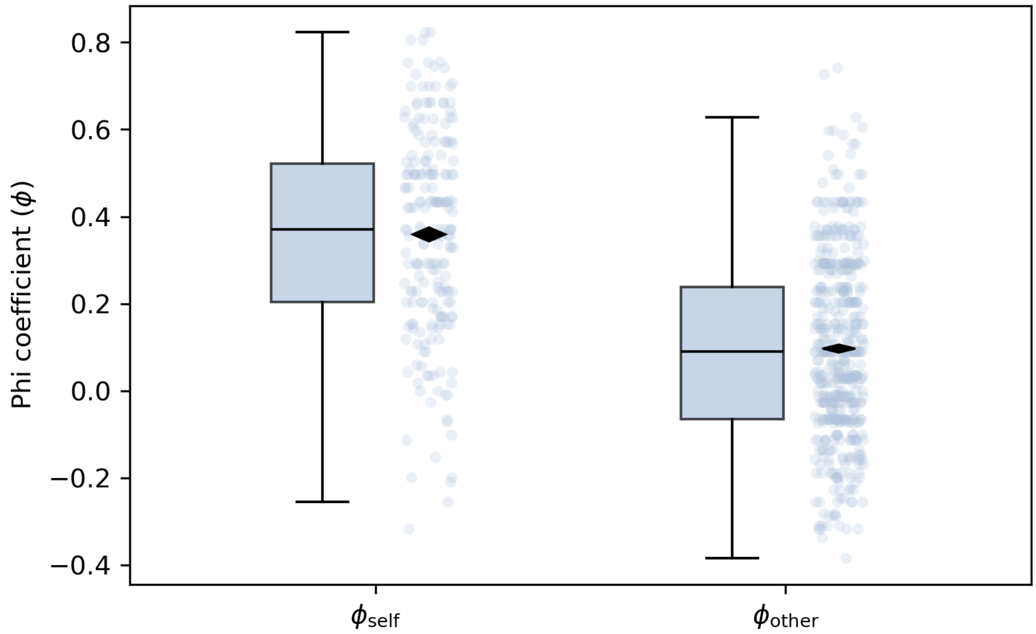


Figure 5: Topic reliability with the selected BERTopic fit. Showing self–self pairs ϕ-coefficient (ϕ_{self}) compared to same–other ϕ-coefficient (ϕ_{other}). For ϕ_{self} all 210 participants are plotted, for ϕ_{other} 500 random pairs are sampled.

Because the ϕ statistic is sensitive to granularity, we also report the *probability of superiority* (also known as the area under the curve, AUC). This is the probability that a randomly sampled self–self similarity score (here ϕ coefficient) is higher than a randomly sampled self–other similarity score (ties counted as half). When we interpret self–self scores as "positives" and self–other scores as "negatives" we can write this as:

$$\mathrm{AUC} = \mathrm{P}(\phi_{\mathrm{self}} > \phi_{\mathrm{other}}) + \frac{1}{2}\mathrm{P}(\phi_{\mathrm{self}} = \phi_{\mathrm{other}}),$$

where ϕ_{self} is a randomly drawn self–self similarity score and ϕ_{other} is a randomly drawn self–other similarity score. We thus measure the probability that a randomly drawn within-person comparison is more similar than a randomly drawn between-person comparison (ties counted as half). An AUC of 0.5 corresponds to chance-level separation, whereas AUC values closer to 1 indicate strong within-person similarity (stability) relative to between-person similarity.

For the selected model, we obtain AUC = .80 with all 10 models falling between $.72 - .82$ (see SI Table 1). As such, based on this relatively coarse-grained measure of semantic similarity, which necessarily introduces some noise (e.g., an outlier topic), we are able to identify the same individual in 80% of randomly drawn pairwise comparisons. This provides a very interpretable indicator that the semantic content of a participant's belief network is reliable across waves as measured by the VN tool.

4.3 Face validity of measured belief networks

Next, we evaluate the face validity of the measured belief networks. As described in Section 2.5, participants rate four different network representations, where each contains the same summary statements (nodes), and only the connections (edges) differ. Since participants rate each network against each other one, we obtain three ratings for each network representation. Here, we focus on whether participants (1) rate their own canvas-created belief network as accurate, and (2) whether they reliably distinguish it from the same network with 50% of the connections randomly flipped (for instance from "supporting" to "conflicting").

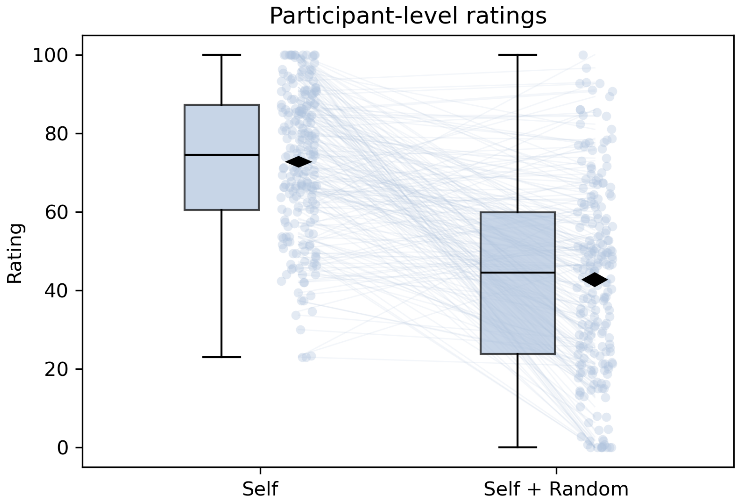


Figure 6: Raw network ratings, where each dot is a participant-level average over three ratings of the same network. Lines connect average ratings of their own canvas-created network (Self) and an identical network with 50% connections flipped (Self + Random).

Average ratings of accuracy for the self-generated canvas network is 72.7 on a 100-point scale, and the average rating for the canvas network with randomly flipped edges is 42.7 (see Figure 6). This shows that participants reliably distinguish their own belief networks from networks with randomly altered connections. We observe a similar discrimination for LLM-extracted networks, which participants rate 66.6 on average, and only 38.2 with 50% randomly flipped connections.

4.4 Concurrent validity of network relationships

To assess the concurrent validity of network relationships, we use a structured pairwise comparison task implemented with an LLM (`gpt-4.1-2025-04-14`) to elicit judgments about specific belief pairs (described in Section 2.6).

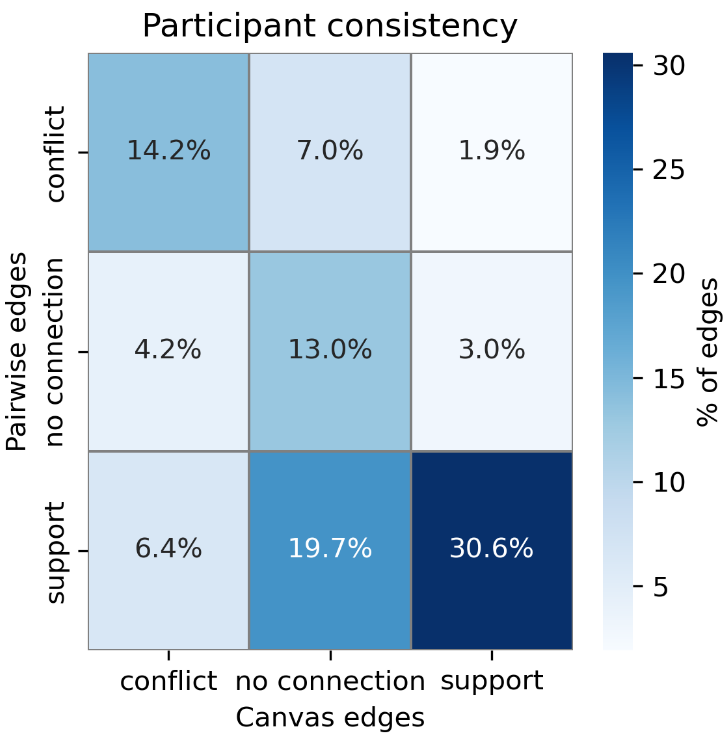


Figure 7: Plot showing the internal consistency of connections created by participants using two different elicitation methods. Percentages are derived by first measuring the percentage in each block at the participant level before aggregating to the population average. Note that this figure includes cases where participants code a connection as both supporting and conflicting in the canvas task. Filtering out these cases makes the results slightly more favorable, but does not change them significantly.

Participants are broadly consistent between the pairwise comparison task and the canvas task, with 84.37% of the beliefs that are coded as either "conflicting" or "supporting" in both tasks having the same sign. Including also ambiguous cases that are either not coded on the canvas or in the pairwise task, 57.8% of all beliefs are coded identically and 8.3% have their sign flipped ("conflict" $\rightarrow$ "support" or "support" $\rightarrow$ "conflict"). The most common inconsistency involves statement-pairs that participants do not code as being connected in the canvas task, but then code as either "supporting" or "conflicting" in the pairwise interview.

4.5 Criterion validity of network structure

The previous analyses establish that there is stability in the semantic content of elicited beliefs. To evaluate criterion validity, we show that network structure predicts *which* topics are more likely to persist at the individual level. A theoretical prediction is that topics that are more interconnected with other topics at the individual level should be more likely to persist across waves, consistent with work showing that higher connectivity of attitude networks is associated with greater stability over time and resistance to change (Dalege, Borsboom, Harreveld, et al. 2017; Dalege, Galesic, and Olsson 2025). We define a connection between two topics as follows: Using BERTopic, we map each statement of a participant to a topic (see Section 4.2). A connection between topic T1 and topic T2 exists if the participant created a connection (supporting or conflicting) between any two statements associated with these topics (including self-loops); the connection weight is the number of connections between statements associated with the corresponding topics.

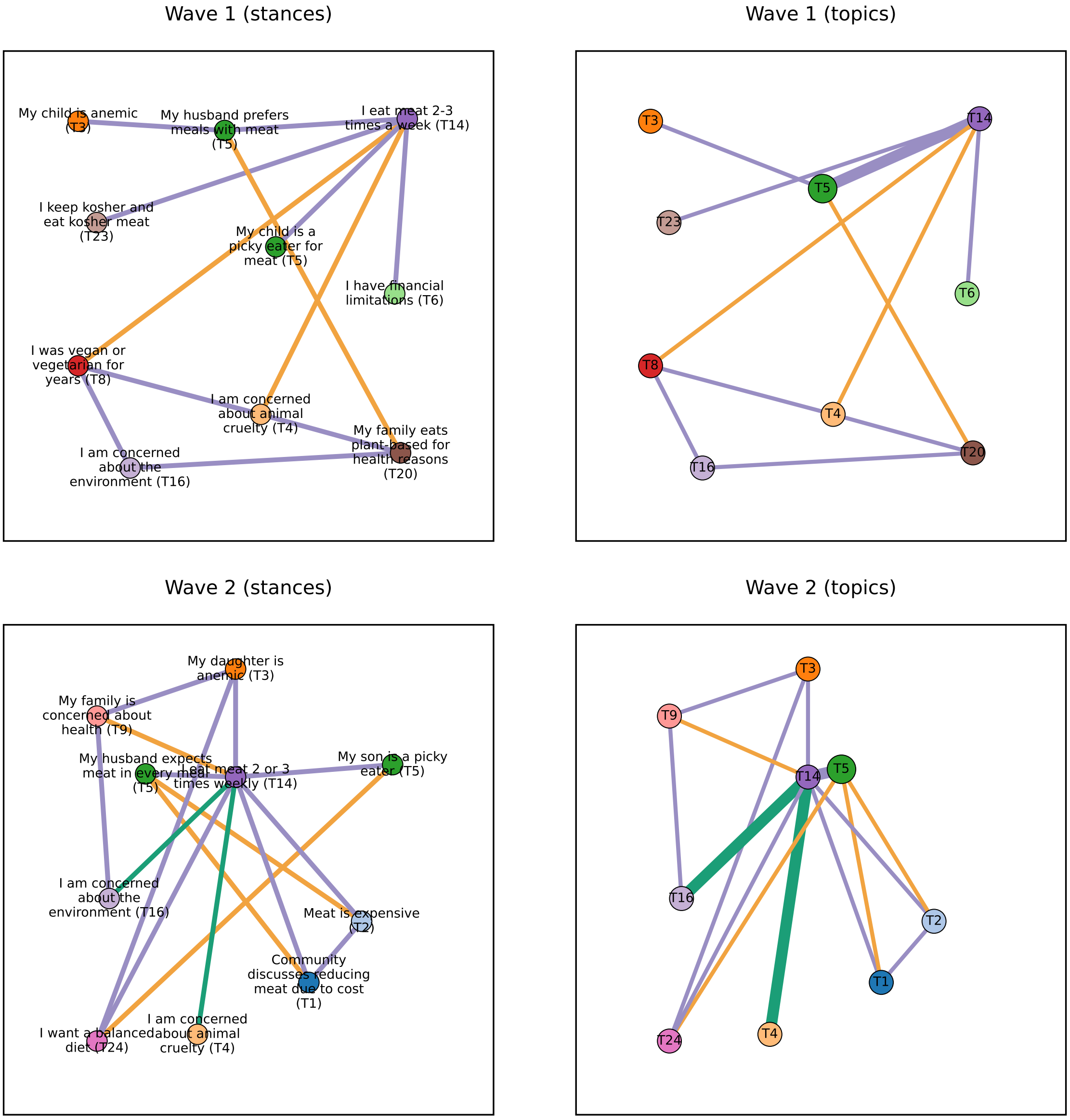


Figure 8: Mapping between canvas networks (left) and corresponding topic assignments (right) for one individual (wave 1 and wave 2). The colors of statements on the left match the topics that they map to on the right. Purple connections are coded by the participant as "supporting", orange connections are coded by the participant as "conflicting", and green connections are shown when the participant has created both types of connections between two beliefs. Placement of statements on the left identical to how the participant positioned them. The thickness of topic connections (right) represents the number of connections between corresponding statements. Similar figures for all 210 participants can be found at `https://github.com/victor-m-p/beliefs-narratives-networks`.

Using topic-connections as the number of links between associated statements, we obtain a degree for each topic. For instance, if a participant had four statements (A, B, C, D) that map to topics (A $\rightarrow$ T1, B $\rightarrow$ T1, C $\rightarrow$ T2, D $\rightarrow$ T3) and the user connected A with B and B with C on the canvas. Then we translate these statement connections into topic connections from topic T1 to itself and from T1 to T2. As such, the degree of topic T1 is 2, of T2 is 1, and of T3 is 0. Many other topics (T4 ...) are not covered by this participant at all. For a real example of a participant-generated network see Figure 8; where statements (A, B, C, ...) are shown on the left and corresponding topics on the right (T1, T2, T3, ...).

In line with the theoretically driven hypothesis, we find that more strongly connected topics in wave 1 (at the individual level) are indeed more likely to appear in wave 2. The observed pattern for the selected BERTopic fit is shown in Figure 9. This general pattern holds across all 10 BERTopic models (both excluding and including the outlier topic).

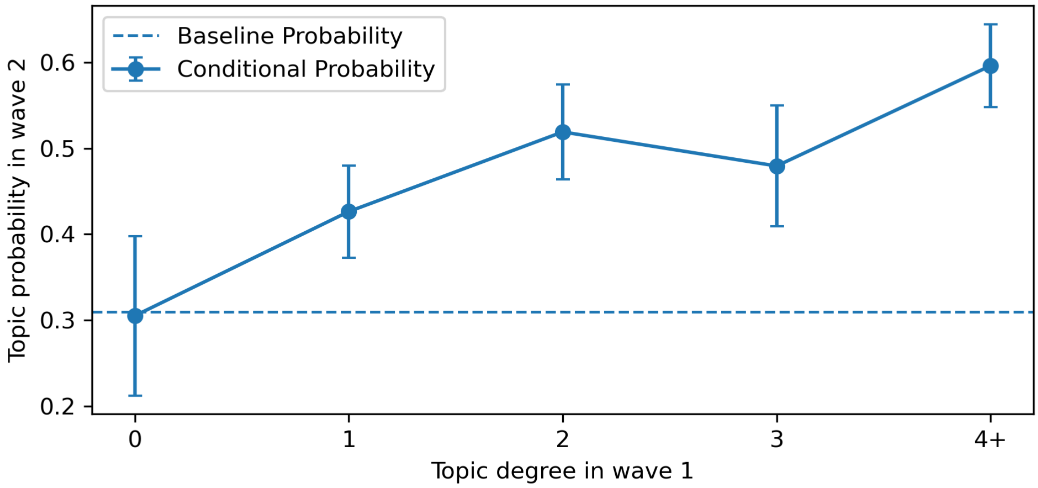


Figure 9: Probability of topic persistence (presence in wave 2) conditional on topic degree in wave 1 at the individual level. The plot uses the selected BERTopic fit and includes the outlier topic. We observe a consistent trend for all 10 BERTopic fit models, both including and excluding outliers. Baseline takes into account that some topics are more common than others.

5 Training

Evaluating "supporting" and "conflicting" relationships between beliefs is not necessarily an intuitive task. Therefore, we asked the participants to complete structured training examples before using the canvas tool. In wave 1, participants completed three vignette-based training tasks; in wave 2, they completed one randomly selected task as a refresher training (see Section 7.1.4).

We design the summary statements for the training examples in such a way that each statement more-or-less is either "supporting" or "conflicting" with the goal described in each vignette (e.g., the statement "Alex practices daily with an app" is supporting the goal to learn a new language). We define a simple consistency rule: statements that are both supporting or both conflicting with the goal in the vignette should themselves be connected by a supporting edge, whereas statements with opposite signs should be connected by a conflicting edge.

We use the degree to which participants obey this pattern as as a basic evaluation of participant consistency, and whether they improve at the task during training. Across all training examples (both wave 1 and wave 2) 87.21% of participant-created connections are consistent with this rule. Participants created "conflicting" connections slightly more consistently with the rule (90.76%) than "supporting" ones (83.71%), likely in part due to the fact that participants always create "supporting" connections before "conflicting" connections, allowing them adjust to the task. The most meaningful gain from training is seen for "supporting" edges which jump from 74.33% for the first training to 86.26% for the second training (same wave) and 90.06% for the third training (same wave). In all cases, we first aggregate at the participant-level; consistency is higher for direct aggregation of all connections across participants.

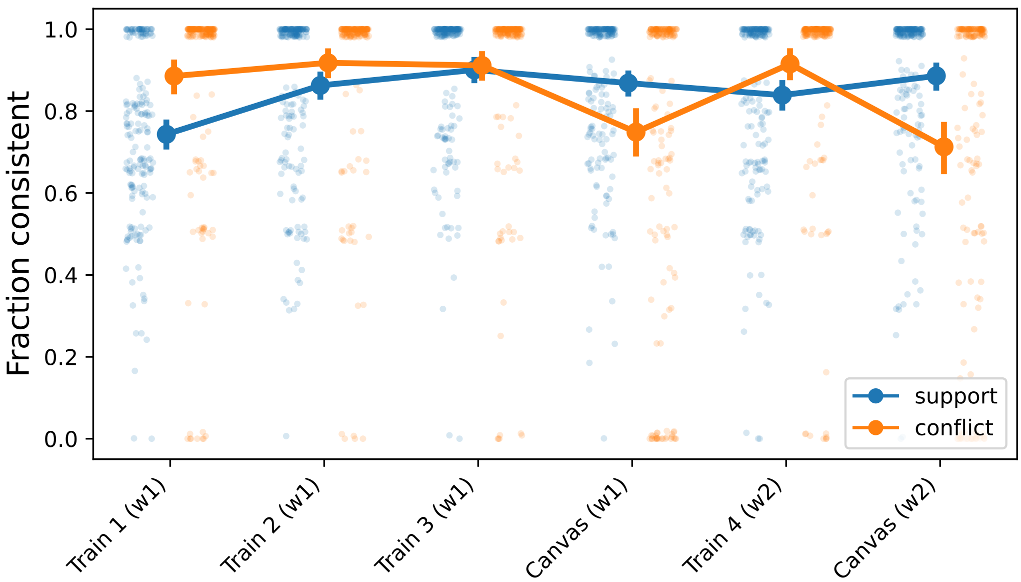


Figure 10: Consistency of individual codings according to the logic explained in the main text. Showing participant-level consistency (i.e., first aggregated by individual and then by population). Shows a positive effect of training on consistency, especially initially for "supporting" connections. Suggests that "conflicting" connections are more difficult for participants to code with their own statements or that they do not obey standard notions of consistency to the same degree as "supporting" connections.

We can apply the same consistency logic to the codings that participants create on the canvas using the summary statements extracted from their interviews. In this case, we rely on the LLM (`gpt-4.1-2025-04-14`) to assign an overall direction to each statement (overall leading to *more* or *less* meat-eating). This is extracted live during the experiment in the same call that extracts summary statements. We observe that consistency for the "supporting" connections remains at the same level between training examples and the canvas task, while "conflicting" connections appear to be harder for participants to code with their own summary statements. It is important to say that some connections could be "true" without obeying this consistency requirement. This analysis aims simply to provide a rough estimate of how consistently we can expect participants to do the task.

6 Discussion

The Visual Network (VN) tool elicits individual belief networks using a combination of LLM-guided semi-structured interview and a visual canvas on which participants indicate relationships between their beliefs extracted from the interviews. The tool does not rely on pre-specified questionnaires that assume the same set of beliefs for all participants, and allows for different relationships at the individual level. Participants report beliefs that are relevant to them within a given domain and indicate how these beliefs are related. We show that basic network properties and semantic content are stable within participants across waves. Participants recognize their own canvas-created belief networks when compared to random networks. To a large extent, belief relations created in the VN tool match the belief relations indicated through direct pairwise evaluation. We show that the individual-level network structure of beliefs is predictive of which topics persist for individuals.

Compared to longitudinal methods for inferring individual belief networks, the VN tool is less cost-intensive, less intrusive, and more suitable for measuring the relations between beliefs that are relatively stable over time–because it does not rely on belief change to measure belief relations. Compared to pairwise comparison methods, it places less cognitive demand on participants while still capturing relationships between beliefs. More fundamentally, the VN tool avoids explicitly asking participants to consider all possible relationships between their beliefs, which might induce reports of previously non-existing connections. In addition, participants do not have to judge each relationship in isolation, but can consider other beliefs when evaluating how beliefs are related.

Tools like the Visual Network tool presented here could allow testing of individual-level hypotheses about belief dynamics that have previously not been feasible. Core assumptions of belief network theories have not been empirically testable given available methods (Brandt and Vallabha 2023). For instance, the prediction of "dynamic constraint"–if one belief changes, then closely connected beliefs should come under pressure as well–has only been evaluated using collective belief networks (Turner-Zwinkels and Brandt 2022). Brandt and Vallabha (2023) has attempted to test theoretical belief network predictions at the individual level using a pairwise conceptual similarity task to measure these networks. While this addresses the gap in the research, their empirical investigation relied on pre-specified and relatively few beliefs to measure individual belief networks, which could miss beliefs that are individually important (Brandt and Vallabha 2023).

In addition, most previous research on belief networks infer belief networks on the population level from correlations between participants (Brandt, Sibley, and Osborne 2019; Boutyline and Vaisey 2017; DellaPosta, Shi, and Macy 2015). Few previous studies have asked participants to directly report how they perceive beliefs, motivations, or behaviors to be related (Brandt and Morgan 2022; Brandt 2022). An advantage of population-based methods is that they can potentially capture implicit associations that might not be available upon introspection, but as discussed above this approach also constrains the types of belief networks that we can study. On the other hand, the ability of participants to accurately introspect about how beliefs relate to each other is not well understood outside of conceptual similarity judgment tasks (Brandt 2022). Our results indicate that when participants are asked about "conflicting" and "supporting" connections they can report these in a relatively reliable way.

The VN tool has several limitations that should be explored further. One limitation is that the networks it captures are for now quite small. For practical reasons, we constrain the maximum number of summary statements (belief nodes) to 10. Furthermore, many participants create only a few connections between their beliefs. This sparsity may reflect genuine features of how they organize their beliefs, but it may also reflect how participants engage with the task or interpret what constitutes a link. When belief networks are sparse, small differences in the number of reported links can have large effects on network measures, making it difficult to separate belief structure from features of the elicitation process. Furthermore, the notions of "support" and "conflict" between different beliefs could be understood differently by different participants. We do observe that the connections that participants create using the VN tool and the pairwise comparison are largely consistent, but participants sometimes only report a connection using one of the elicitation methods, and they occasionally report connections in opposing directions. Finally, the present study uses a commercial LLM and focuses on meat-eating, but our ongoing research shows that the VN tool can be successfully implemented with commercial or open-source LLMs, and used to elicit beliefs about different topics.

References

Bartels, Larry M (1999). "Panel effects in the American national election studies". In: *Political Analysis* 8.1, pp. 1–20.

Beardsworth, Alan and Alan Bryman (2004). "Meat consumption and meat avoidance among young people: An 11-year longitudinal study". In: *British Food Journal* 106.4, pp. 313–327.

Bizyaeva, Anastasia, Alessio Franci, and Naomi Ehrich Leonard (2022). "Sustained oscillations in multi-topic belief dynamics over signed networks". In: *arXiv preprint arXiv:2210.00353*.

Boutyline, Andrei and Stephen Vaisey (2017). "Belief network analysis: A relational approach to understanding the structure of attitudes". In: *American journal of sociology* 122.5, pp. 1371–1447.

Brandt, Mark J (2022). "Measuring the belief system of a person." In: *Journal of Personality and Social Psychology* 123.4, p. 830.

Brandt, Mark J and G Scott Morgan (2022). "Between-person methods provide limited insight about within-person belief systems." In: *Journal of Personality and Social Psychology* 123.3, p. 621.

Brandt, Mark J, Chris G Sibley, and Danny Osborne (2019). "What is central to political belief system networks?" In: *Personality and Social Psychology Bulletin* 45.9, pp. 1352–1364.

Brandt, Mark J and Willem WA Sleegers (2021). "Evaluating belief system networks as a theory of political belief system dynamics". In: *Personality and Social Psychology Review* 25.2, pp. 159–185.

Brandt, Mark J and Shree Vallabha (2023). "Inter-attitude centrality does not appear to reduce persuasion for political attitudes". In: *European Journal of Social Psychology* 53.7, pp. 1342–1358.

Chambon, Monique, Jonas Dalege, Denny Borsboom, et al. (2023). "How compliance with behavioural measures during the initial phase of a pandemic develops over time: A longitudinal COVID-19 study". In: *British Journal of Social Psychology* 62.1, pp. 302–321.

Chambon, Monique, Jonas Dalege, Lourens J Waldorp, et al. (2022). "Tailored interventions into broad attitude networks towards the COVID-19 pandemic". In: *PloS one* 17.10, e0276439.

Cook, Thomas D, Donald Thomas Campbell, and William Shadish (2002). *Experimental and quasi-experimental designs for generalized causal inference*. Vol. 1195. Houghton Mifflin Boston, MA.

Dalege, Jonas, Denny Borsboom, Frenk van Harreveld, et al. (2017). "Network structure explains the impact of attitudes on voting decisions". In: *Scientific reports* 7.1, p. 4909.

Dalege, Jonas, Denny Borsboom, Frenk Van Harreveld, et al. (2016). "Toward a formalized account of attitudes: The Causal Attitude Network (CAN) model." In: *Psychological review* 123.1, p. 2.

Dalege, Jonas and Tamara van der Does (2022). "Using a cognitive network model of moral and social beliefs to explain belief change". In: *Science advances* 8.33, eabm0137.

Dalege, Jonas, Mirta Galesic, and Henrik Olsson (2025). "Networks of beliefs: An integrative theory of individual- and social-level belief dynamics". In: *Psychological Review* 132.2, pp. 253–290. DOI: 10.1037/rev0000494.

DellaPosta, Daniel, Yongren Shi, and Michael Macy (2015). "Why do liberals drink lattes?" In: *American Journal of Sociology* 120.5, pp. 1473–1511.

DeVellis, Robert F and Carolyn T Thorpe (2021). *Scale development: Theory and applications*. Sage publications.

Goldberg, Amir and Sarah K Stein (2018). "Beyond social contagion: Associative diffusion and the emergence of cultural variation". In: *American Sociological Review* 83.5, pp. 897–932.

Grootendorst, Maarten (2022). "BERTopic: Neural topic modeling with a class-based TF-IDF procedure". In: *arXiv preprint arXiv:2203.05794*.

Ha, Seungwoong et al. (2025). "Dynamics of collective minds in online communities". In: *arXiv preprint arXiv:2504.08152*.

Hasan, Khaerullah Fadhli Arasy et al. (2025). "What is the Most Important Issue in Indonesia's Political Belief System? Network Analysis of Attitude and Voting Behavior in 2014 and 2019". In: *Makara Human Behavior Studies in Asia* 29.2, pp. 115–136.

Hopwood, Christopher J et al. (2020). "Health, environmental, and animal rights motives for vegetarian eating". In: *PloS one* 15.4, e0230609.

Hout, Michael C, Stephen D Goldinger, and Ryan W Ferguson (2013). "The versatility of SpAM: a fast, efficient, spatial method of data collection for multidimensional scaling." In: *Journal of Experimental Psychology: General* 142.1, p. 256.

Kiley, Kevin and Stephen Vaisey (2020). "Measuring stability and change in personal culture using panel data". In: *American Sociological Review* 85.3, pp. 477–506.

Mansueto, Alessandra C et al. (2023). "Investigating the feasibility of idiographic network models." In: *Psychological methods* 28.5, p. 1052.

Moulavi, Davoud et al. (2014). "Density-based clustering validation". In: *Proceedings of the 2014 SIAM international conference on data mining*. SIAM, pp. 839–847.

OpenAI (2025a). *GPT-4.1 model — OpenAI API Documentation*. URL: https://platform.openai.com/docs/models/gpt-4.1 (visited on 11/13/2025).

— (Apr. 2025b). *Introducing GPT-4.1 in the API*. URL: https://openai.com/index/gpt-4-1/ (visited on 11/13/2025).

OpenResearch (Aug. 2025a). *Can Conversational AI Improve Survey Research? Early Insights from Testing a Chat-Based Survey*. https://www.openresearchlab.org/findings/ai-surveys. Accessed: 2025-11-13.

— (2025b). *Implementation Details: AI-Enhanced Survey System*. https://www.openresearchlab.org/research/implementation-details. Accessed: 2025-11-13.

Park, Joon Sung et al. (2024). "Generative agent simulations of 1,000 people". In: *arXiv preprint arXiv:2411.10109*.

Piazza, Jared et al. (2015). "Rationalizing meat consumption. The 4Ns". In: *Appetite* 91, pp. 114–128.

Reimers, Nils and Iryna Gurevych (2019). "Sentence-bert: Sentence embeddings using siamese bert-networks". In: *arXiv preprint arXiv:1908.10084*.

Rodriguez, Nathaniel, Johan Bollen, and Yong-Yeol Ahn (2016). "Collective dynamics of belief evolution under cognitive coherence and social conformity". In: *PLoS one* 11.11, e0165910.

Rothgerber, Hank and Daniel L Rosenfeld (2021). "Meat-related cognitive dissonance: The social psychology of eating animals". In: *Social and Personality Psychology Compass* 15.5, e12592.

Scheffer, Marten et al. (2022). "Belief traps: Tackling the inertia of harmful beliefs". In: *Proceedings of the National Academy of Sciences* 119.32, e2203149119.
Schwarz, Norbert (1999). "Self-reports: How the questions shape the answers." In: *American psychologist* 54.2, p. 93.
Schweighofer, Simon, Frank Schweitzer, and David Garcia (2020). "A weighted balance model of opinion hyperpolarization". In.
Tourangeau, Roger, Lance J Rips, and Kenneth Rasinski (2000). "The psychology of survey response". In.
Turner-Zwinkels, Felicity M and Mark J Brandt (2022). "Belief system networks can be used to predict where to expect dynamic constraint". In: *Journal of Experimental Social Psychology* 100, p. 104279.
Turner-Zwinkels, Felicity M, Branden B Johnson, et al. (2021). "Conservatives' moral foundations are more densely connected than liberals' moral foundations". In: *Personality and Social Psychology Bulletin* 47.2, pp. 167–184.
Velez, Yamil, Patrick Liu, and Scott Clifford (2025). "When Information Affects Attitudes: The Effectiveness of Targeting Attitude-Relevant Beliefs". In.
Vlasceanu, Madalina, Ari M Dyckovsky, and Alin Coman (2024). "A network approach to investigate the dynamics of individual and collective beliefs: Advances and applications of the bending model". In: *Perspectives on Psychological Science* 19.2, pp. 444–453.
Xiao, Shitao et al. (2024). "C-pack: Packed resources for general chinese embeddings". In: *Proceedings of the 47th international ACM SIGIR conference on research and development in information retrieval*, pp. 641–649.
Zhang, Yanzhao et al. (2025). "Qwen3 Embedding: Advancing Text Embedding and Reranking Through Foundation Models". In: *arXiv preprint arXiv:2506.05176*.

7 Supplementary Information

For code and materials, see `https://github.com/victor-m-p/beliefs-narratives-networks`.

7.1 Survey Tool Description

In this section, we document one complete run of the survey instrument completed on 2026-03-04 by the lead author. A full video of this completion, along with code to deploy the VN tool, and supporting scripts, can be found at `https://github.com/victor-m-p/beliefs-narratives-networks`.

7.1.1 Semi-structured LLM interview

As described in Section 2.1, the first major task for participants is to complete an 8-round semi-structured interview with an LLM-interviewer. We cannot share the raw participant interviews here, so to give an impression, the interview conducted by the lead author on 2026-03-04 is shown below along with a screenshot:

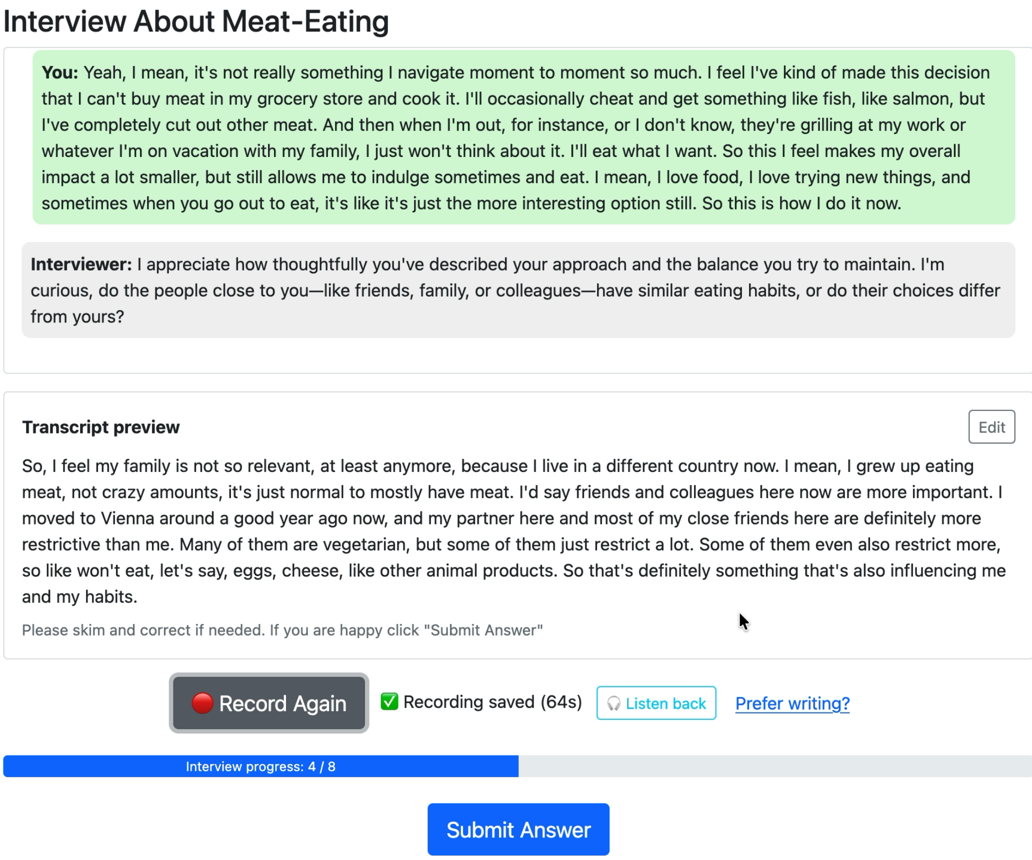


Figure 11: Screenshot from survey completed by the lead author on 2026-03-04. Here showing the semi-structured interview about meat-eating with the LLM. In green a previous utterance by the lead author, in grey the follow-up question by the LLM, and in white ("Transcript preview") the transcription (`whisper`) of the answer to the present question.

Interviewer: In this interview, we're talking about meat-eating habits. Is this something you've thought much about, or not really?
Participant: It's something I've thought quite a bit about, I mean now I also read about it and do research about it, but even before that it was, I mean it's come up a few times, especially when I've moved in with people, let's say that are vegetarian or just restrict meat, when I've gotten to know people very well that have different habits, yes it's definitely something that's been coming up a few times and also where I've changed my own habits a bit and have read about factory farming practices, etc. Yes I would say it's something that is like a salient topic in my circles and is something I'm also navigating myself.

Interviewer: Thanks for sharing that–it's interesting to hear how your habits and thoughts have evolved, especially as you've learned more and spent time with people who have different approaches. Could you tell me a bit about your own current meat-eating habits?
Participant: Yeah, so my own current meat-eating habits are, I mean, I still eat meat, right, but I think what I settled on for now is just that I don't eat meat regularly, I don't buy meat and cook it myself, but I'll allow myself to eat meat if it's offered, if I'm out, if it's interesting, but I mostly just try to not buy cheap meat, where I know that the animals have suffered, so that's kind of the compromise I've

arrived on for now.

Interviewer: Thanks for explaining the compromise you've found with your meat-eating–it's interesting how you balance your choices based on circumstances and concerns about animal welfare. Could you share a bit more about what motivates you personally to eat meat on some occasions, or to avoid it on others?
Participant: Yeah, I mean, it's not really something I navigate moment to moment so much. I feel I've kind of made this decision that I can't buy meat in my grocery store and cook it. I'll occasionally cheat and get something like fish, like salmon, but I've completely cut out other meat. And then when I'm out, for instance, or I don't know, they're grilling at my work or whatever I'm on vacation with my family, I just won't think about it. I'll eat what I want. So this I feel makes my overall impact a lot smaller, but still allows me to indulge sometimes and eat. I mean, I love food, I love trying new things, and sometimes when you go out to eat, it's like it's just the more interesting option still. So this is how I do it now.

Interviewer: I appreciate how thoughtfully you've described your approach and the balance you try to maintain. I'm curious, do the people close to you–like friends, family, or colleagues–have similar eating habits, or do their choices differ from yours?
Participant: So, I feel my family is not so relevant, at least anymore, because I live in a different country now. I mean, I grew up eating meat, not crazy amounts, it's just normal to mostly have meat. I'd say friends and colleagues here now are more important. I moved to Vienna around a good year ago now, and my partner here and most of my close friends here are definitely more restrictive than me. Many of them are vegetarian, but some of them just restrict a lot. Some of them even also restrict more, so like won't eat, let's say, eggs, cheese, like other animal products. So that's definitely something that's also influencing me and my habits.

Interviewer: Thanks for explaining how your friends and partner's habits influence you. I'm curious, do you have a sense of what motivates them to be more restrictive with meat or other animal products?
Participant: Yeah, I think it differs a bit. I do think, I mean, some of them made this decision many years ago, right, to just cut meat out completely. Yeah, I mean, I think some of them, many of my close friends here are very concerned about climate change and environmental impact. I think also some of them are very concerned about animal welfare, which I think, I mean, that's definitely my bigger concern with this. Then I think there's like smaller differences, right? I think I'm mostly concerned about animal suffering. I won't feel so bad about eating meat if I know that the animal has lived a decent life. But I think some of them are just fundamentally against, let's say, any kind of farming for meat. I guess that's sort of philosophically a little bit more, I don't know, Kantian, if I'm like a utilitarian, but maybe that's like overthinking it. I mean, I also just think it's kind of a part of a left-wing identity. Many of my friends here are more active on the left wing, also organizing, protesting, doing different things. And I think being a vegetarian is definitely like kind of that package in a way.

Interviewer: Thanks for sharing those reflections about your friends' and your own motivations–it's really interesting to hear how different concerns like animal

welfare, the environment, or philosophical beliefs factor in. You mentioned that being vegetarian can also be part of a broader identity or community for some people–could you say a bit more about how you see that playing out among your friends or in your social circles?
Participant: Yeah, I'm not sure how much more I can elaborate on that. I just think that most people in my circle, me included, do hold the belief that eating less meat is the more moral and ethical choice for lots of reasons. It just is true that it has much less environmental impact, and it's also true that many animals suffer greatly in factory farming. I actually don't think that my friends and my partner are judgmental. I think they're pretty chill, and most of them also will occasionally cheat and try meat, maybe not all of them. So I think they are also just showing by example that this is a choice you can make, and yeah, I think that is also just inspiring me to reconsider my own choice and how I go about this.

Interviewer: Thanks for sharing how your friends and partner influence you and how shared beliefs shape your group's approach–it's interesting to hear about the positive, non-judgmental way this plays out. Looking back at your journey, has there been a particular moment or experience that really shifted the way you think about eating meat?
Participant: Good question. I mean, it's a little bit difficult to pinpoint one moment. I think there's been a couple of different things. I mean, I do think as many others, one thing that has shocked me has been seeing clips and videos from factory farms. I think for many people, and rightly so, this is a shocking experience. I also a few years ago read Peter Singer's Animal Liberation, which sort of documents how terrible factory farming is. And so the scale of suffering involved from a very utilitarian perspective, but it was still a very impactful book for me to read. I mean, to be fair, I've also been convinced for many years that I should eat less meat, and I've still maintained eating meat, right, because of various things. Because I'm not entirely self-consistent, I enjoy eating it, etc. So I think maybe what has more driven me to start changing has been, let's say, the first time a few years ago when I moved in to a collective with vegetarians. That's definitely when my day-to-day habits started changing and maybe more coming into alignment with my beliefs.

Interviewer: Thank you for sharing so much about your experiences and reflections–it's been really insightful. Before we wrap up, is there anything else about your thoughts, experiences, or journey around eating meat that you'd like to mention that we haven't covered yet?
Participant: Yeah, I mean, I guess, I mean, one thing we haven't talked so much about is why I still do eat meat sometimes. I think we've talked a lot about all of my motivations, both social and moral, to cut down on meat. I do think there is, I mean, I greatly respect the people that have made a more consistent sort of choice than I have, to really just cut out meat completely. And that might also be a little bit easier than having to evaluate from moment to moment whether you should allow yourself, but I also think that a lot of people could cut down 50, 60, 80% of their meat intake without experiencing much, quote-unquote, disruption. I definitely feel that so far I haven't made any difficult decisions. It's very easy for me, maybe not for everyone, I also respect that, but it's easy for me to cook great vegetarian food myself and then I still allow myself to experience sometimes great tasting meat when I'm out, and to not turn down food, to eat whatever is offered.

> So right now I'm actually pretty happy with having reduced so much, but still sort of, yeah, not having restricted myself so much, but let's see.

After completing the LLM-based interview, participants answer a few questions that help us assess the quality and participant experience (see Figure 2). All of the questions below are answered on a scale from 1 to 100 with dynamically changing anchors.

1. "How was your overall experience with the conversational survey?" (Excellent, Good, Average / Neutral, Not good, Terrible) (OpenResearch 2025a)
2. "The questions in the interview were relevant" (Strongly agree, Somewhat agree, Neither agree nor disagree, Somewhat disagree, Strongly disagree) (OpenResearch 2025a).
3. "It was easy to express my thoughts in the chat format" (Strongly agree, Somewhat agree, Neither agree nor disagree, Somewhat disagree, Strongly disagree) (OpenResearch 2025a).
4. "I felt comfortable enough during the conversation to honestly describe my habits and motivations" (Strongly agree, Somewhat agree, Neither agree nor disagree, Somewhat disagree, Strongly disagree) (OpenResearch 2025a)
5. "Interacting with the AI model felt creepy or intrusive" (Strongly agree, Somewhat agree, Neither agree nor disagree, Somewhat disagree, Strongly disagree) (OpenResearch 2025a).

7.1.2 Summary Statements

As described in Section 2.2, the LLM extracts up to 10 summary statements based on the interview transcript (see above). Participants rate how *accurate* and *relevant* each statement is, including two attention checks. Based on the interview with the lead author on 2026-03-04 the following 10 statements were extracted:

1. I do not buy meat to cook
2. I eat meat when eating out
3. I avoid cheap meat
4. I love trying new foods
5. I am concerned about animal suffering
6. I am concerned about environmental impact
7. My social circle restricts animal products
8. Social contacts are concerned about animal welfare
9. Videos of factory farming shocked me
10. Reading Animal Liberation influenced me

And presented in random order along with the two attention checks: (1) "I eat beans to get enough cholesterol", and (2) "My friends often go out to eat seahorse".

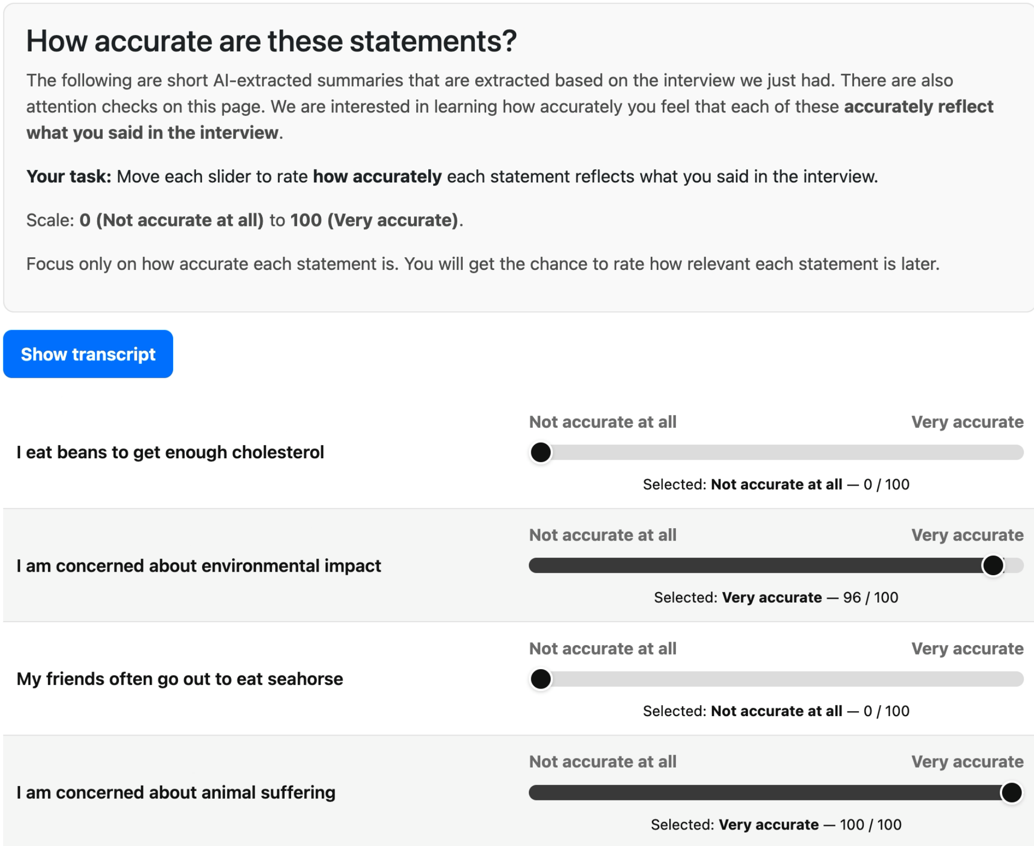


Figure 12: Screenshot from survey completed by the lead author on 2026-03-04. Showing the page where participants rate the *accuracy* of summary statements extracted by the LLM. In this screenshot we see ratings of two attention checks: "I eat beans to get enough cholesterol" and "My friends often go out to eat seahorse" along with two LLM-extracted summaries based on the interview transcript (the participant scrolls down to rate all statements on one page). The interface for the *relevance* ratings of summary statements is identical.

We observe consistently high ratings on both *accuracy* and *relevance* for LLM-extracted summary statements across both waves, and consistently low ratings on both *accuracy* and *relevance* for the two attention checks (see Figure 3). The specific questions answered are:

1. "Rate how **accurately** each statement reflects what you said in the interview" (100-point slider with 7-point Likert annotations, anchored at "Not accurate at all", "Very accurate")
2. "Rate how **relevant** each statement is for your personal meat-eating habits" (100-point slider with 7-point Likert annotations, anchored at "Not relevant at all", "Very relevant").

7.1.3 Canvas task

As described in Section 2.3, the canvas task consists of 3 screens (spatial arrangement, "supporting" connections, and "conflicting" connections). Because the lead author rated all 10 summary statements as at least "Accurate" (60 or above on a 100-point scale) all 10 statements are used here.

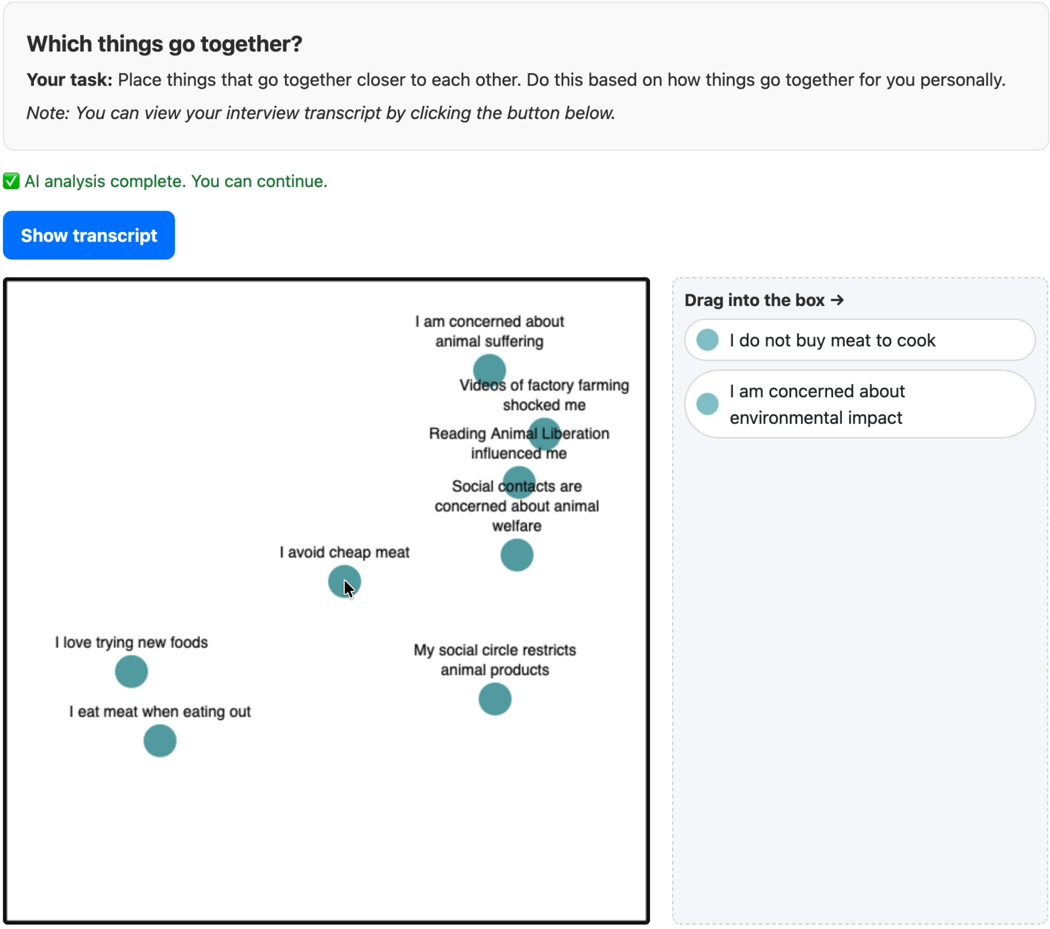


Figure 13: Screenshot from survey completed by the lead author on 2026-03-04. Showing the initial canvas task (spatial arrangement) of summary statements. Initially, the summary statements appear in random order on the right ("Drag into the box") and the participant can proceed when LLM background analysis is done and the participant has placed all statements within the canvas on the left.

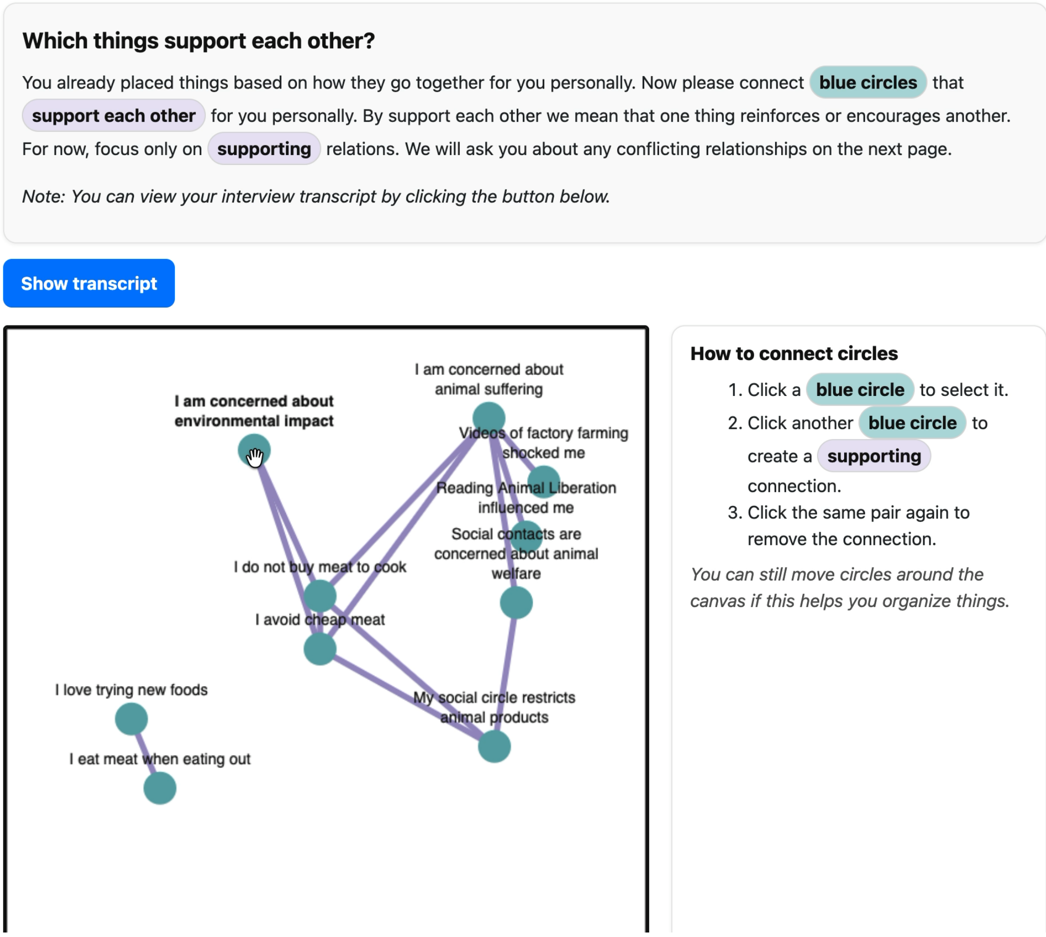


Figure 14: Screenshot from survey completed by the lead author on 2026-03-04. Showing the second canvas task, creating "supporting" connections.

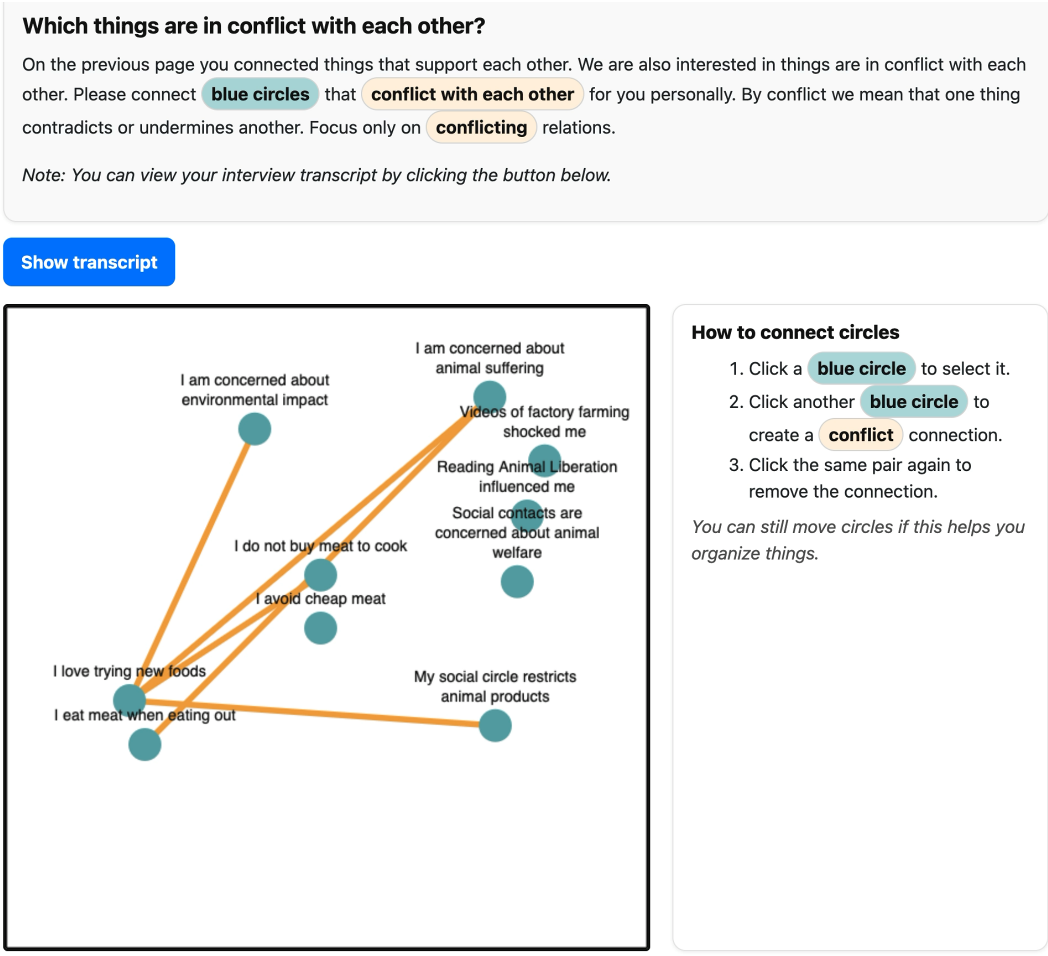
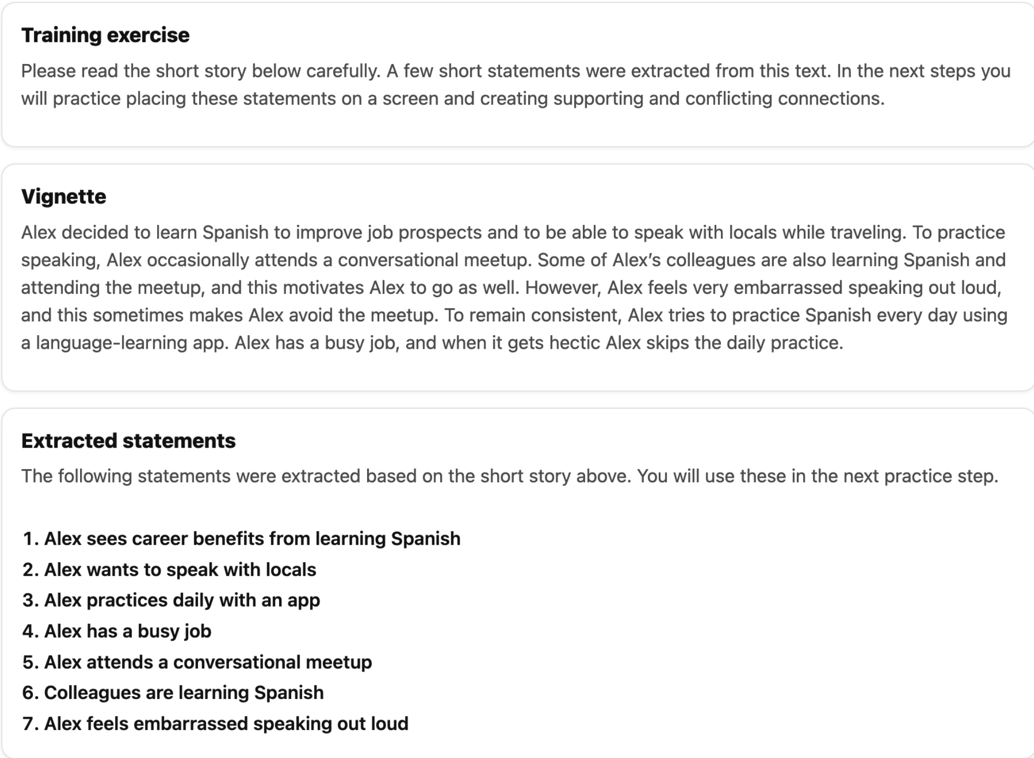


Figure 15: Screenshot from survey completed by the lead author on 2026-03-04. Showing the second canvas task, creating "conflicting" connections.

7.1.4 Training

As described in Section 2.4 participants complete a training exercise before the main canvas task in both wave 1 (3 training examples) and in wave 2 (1 training example). Logically and visually, this follows the structure of the canvas task described above. First, participants visually arrange statements, then they create "supporting" connections, and then they create "conflicting" connections.



Training exercise

Please read the short story below carefully. A few short statements were extracted from this text. In the next steps you will practice placing these statements on a screen and creating supporting and conflicting connections.

Vignette

Alex decided to learn Spanish to improve job prospects and to be able to speak with locals while traveling. To practice speaking, Alex occasionally attends a conversational meetup. Some of Alex's colleagues are also learning Spanish and attending the meetup, and this motivates Alex to go as well. However, Alex feels very embarrassed speaking out loud, and this sometimes makes Alex avoid the meetup. To remain consistent, Alex tries to practice Spanish every day using a language-learning app. Alex has a busy job, and when it gets hectic Alex skips the daily practice.

Extracted statements

The following statements were extracted based on the short story above. You will use these in the next practice step.

1. **Alex sees career benefits from learning Spanish**
2. **Alex wants to speak with locals**
3. **Alex practices daily with an app**
4. **Alex has a busy job**
5. **Alex attends a conversational meetup**
6. **Colleagues are learning Spanish**
7. **Alex feels embarrassed speaking out loud**

Figure 16: Screenshot from survey completed by the lead author on 2026-03-04. Participants see this information screen before proceeding to the spatial arrangement task.

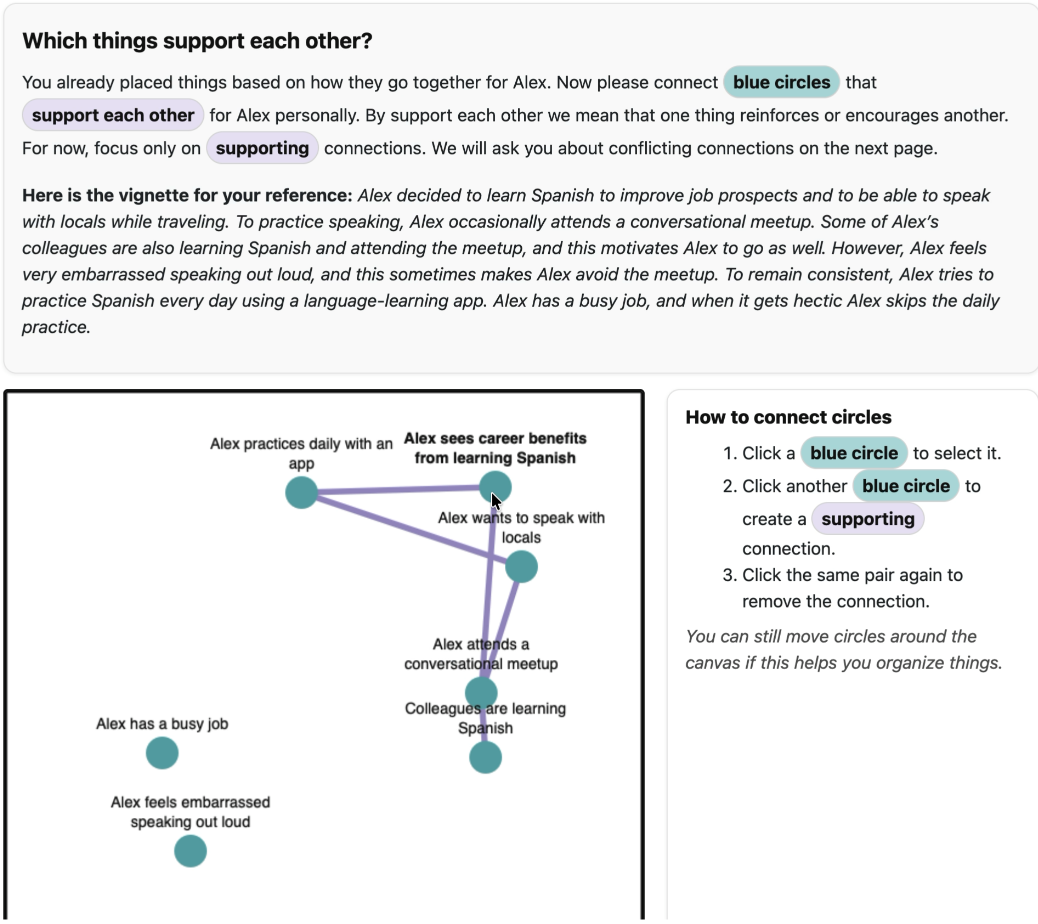


Figure 17: Screenshot from survey completed by the lead author on 2026-03-04. Here showing the creation of "supporting" connections for the "Alex" training scenario. The page for "conflicting" connections is visually identical.

We design 3 training examples (see below), which all take the same basic form. A text with some goal is provided (e.g., learning Spanish) and 7 summary statements are provided, encompassing both the goal, motivations and challenges. In wave 1 each participant completes all 3 training examples in random order and in wave 2 each participant completes 1 training example sampled randomly. We provide live feedback to participants and calculate the consistency of participant codings during the training phase (used in Section 5). This is allowed by design since some pairs of statements are explicitly either in "conflict" or "supportive".

Vignette 1 ("Alex")
Alex decided to learn Spanish to improve job prospects and to be able to speak with locals while traveling. To practice speaking, Alex occasionally attends a conversational meetup. Some of Alex's colleagues are also learning Spanish and attending the meetup, and this motivates Alex to go as well. However, Alex feels very embarrassed speaking out loud, and this sometimes makes Alex avoid the meetup. To remain consistent, Alex tries to practice Spanish every day using a language-learning app. Alex has a busy job, and when it gets hectic Alex skips the daily practice.

1. Alex sees career benefits from learning Spanish
2. Alex wants to speak with locals

3. Alex practices daily with an app
4. Alex has a busy job
5. Alex attends a conversational meetup
6. Colleagues are learning Spanish
7. Alex feels embarrassed speaking out loud

Vignette 2 ("Jordan")

Jordan wants to improve their physical fitness and to have more energy during the day. A friend has invited Jordan to join their gym sessions, and the gym is located on Jordan's way home from work, which makes it convenient to go. Jordan plans to go to the gym three times a week. However, Jordan often feels exhausted after work and feels self-conscious exercising in front of others, and these feelings sometimes lead Jordan to skip the gym.

1. Jordan wants to improve their physical fitness
2. Jordan wants to have more energy during the day
3. Jordan plans to go to the gym three times a week
4. Jordan often feels exhausted after work
5. A friend has invited Jordan to join their gym sessions
6. The gym is located on Jordan's way home from work
7. Jordan feels self-conscious exercising in front of others

Vignette 3 ("Riley"):

Riley wants to get more involved in the local community and registered for a weekly volunteer shift. A close friend also volunteers there, which makes Riley feel welcome. The community center is located close to Riley's home, so getting there is usually easy. However, Riley sometimes needs to take care of a younger relative on short notice, which makes it difficult to attend the volunteer shift consistently. Riley tries to plan ahead each week, using reminders and a shared calendar, but unexpected caregiving needs sometimes interfere.

1. Riley wants to contribute to the local community
2. Riley registered for a weekly volunteer shift
3. A close friend also volunteers there
4. The community center is close to Riley's home
5. Riley sometimes needs to take care of a younger relative on short notice
6. Riley tries to plan ahead using reminders and a shared calendar
7. Unexpected caregiving needs sometimes interfere with Riley's plans

7.1.5 Network Comparison

As described in Section 2.5 we implement (for wave 2) a network comparison task to establish face validity. This consists of 6 pages comparing 4 networks (including LLM-extracted belief networks). See Figure 18.

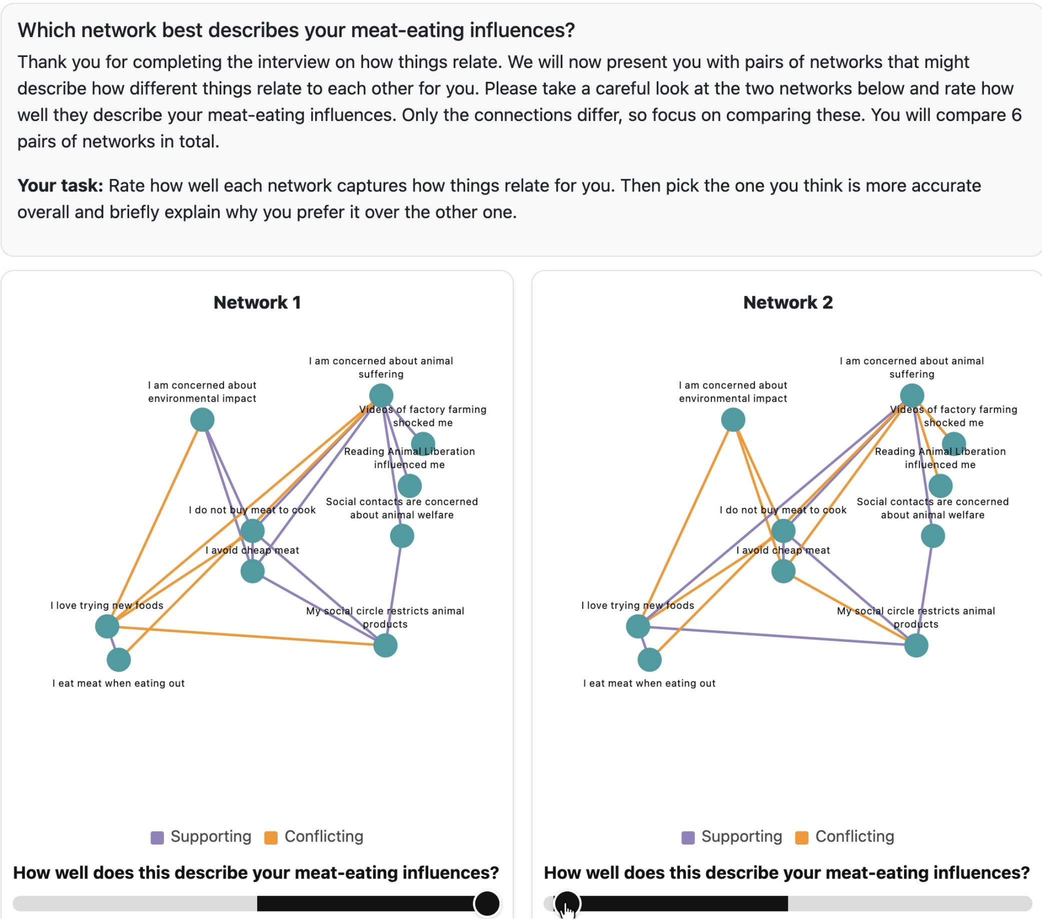


Figure 18: Screenshot from survey completed by the lead author on 2026-03-04. On the left, the actual network of “conflicting” and “supporting” connections created by the lead author on the canvas. On the right, the same nodes but 50% of the connections with the opposite direction.

7.1.6 Pairwise Interview

As described in Section 2.6 we implement (for wave 2) a pairwise comparison task to assess consistency between various elicitation methods for belief connections. The pairwise interview works broadly like the semi-structured interview described above, but combines (1) an open-ended question about a pair of statements, and (2) a closed-ended question which forces the participant to choose a classification. See Figure 19.

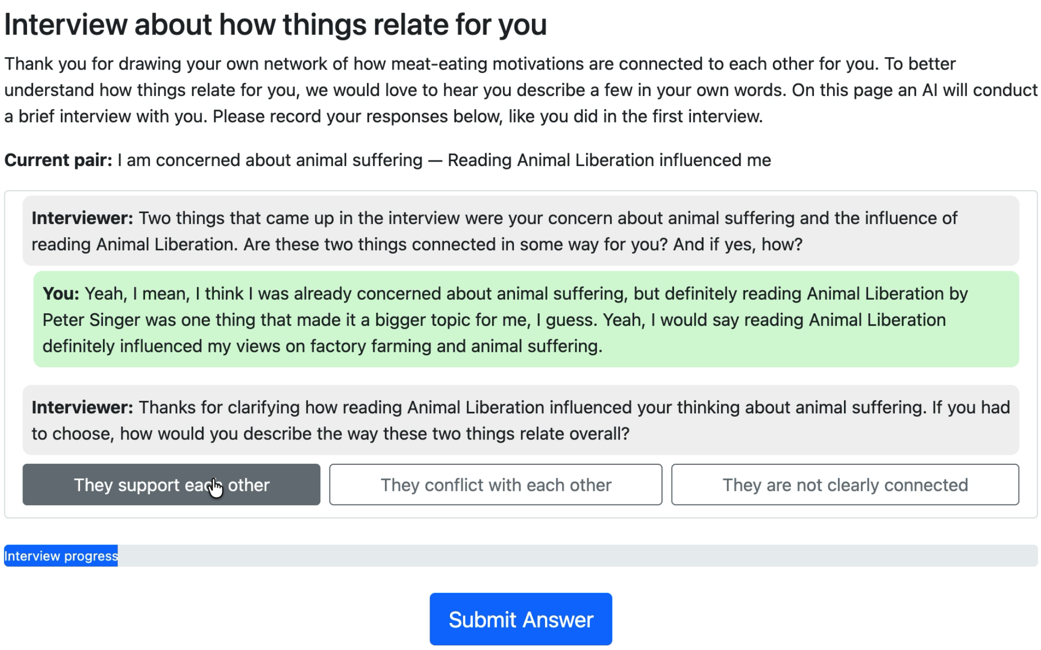


Figure 19: Screenshot from survey completed by the lead author on 2026-03-04. Showing an LLM question, a participant answer, and the follow-up question with three options: "They support each other", "They conflict with each other", and "They are not clearly connected".

7.2 LLM usage

We use Large Language Models (LLMs) for 3 main tasks within the survey instrument. (1) As an AI-interviewer conducting a semi-structured interview about meat-eating, (2) to extract summary statements from an interview transcript about meat-eating, (3) to extract relationships (networks) between these summary statements. In all cases, we use OpenAI's Chat Completions API. Specifically, we use `gpt-4.1-2025-04-14` with temperature set to 0.7. `gpt-4.1` is the smartest non-reasoning model from OpenAI (OpenAI 2025a) which is ideal for our use-cases since we require the best possible understanding combined while ensuring low latency. Comparable work using LLMs as interviewers, to summarize attitudes, and to create personalized arguments, have mostly relied on the `gpt-4o` model from OpenAI (Park et al. 2024; OpenResearch 2025b; Velez, P. Liu, and Clifford 2025). We use the newer `gpt-4.1` model because it beats the older `gpt-4o` model across all relevant benchmarks (OpenAI 2025b).

7.3 Semantic Analysis

First, we select 4 candidate sentence embedding models. `all-MiniLM-L6-v2` and `bert-base-nli-mean-tokens` (Reimers and Gurevych 2019) are historical baselines that have been used in thousands of publications. `bge-large-en-v1.5` (Xiao et al. 2024) and `qwen3-embedding-0.6b` (Y. Zhang et al. 2025) are more recent and powerful sentence embedding models. All models are run through the `SentenceTransformers` framework from `HuggingFace`.

We then run a grid search over hyper-parameters to arrive at optimal settings for the BERTopic model. The most important hyper-parameters are the number of neighbors (neighbors) in UMAP and the minimum cluster size (cluster) for HDBSCAN. We also vary the number of components in UMAP (components). In total, our grid searches over *neighbors* $\in \{15, 30, 45\}$,

$cluster \in \{25, 50, 75\}$, and $components \in \{2, 10\}$.

This results in a total of $3 \times 3 \times 2 \times 4 = 72$ BERTopic fits. We use the following selection criteria to arrive at the final model(s) that we base the analysis reported in Section 4.2 on. (1) First, we remove all BERTopic fits that have more than 40% of all stances assigned to the outlier topic (-1) and the 0-topic. This removes 11 BERTopic fits and leaves us with 61 remaining. Then we remove all BERTopic fits that have less than 5 topics and more than 40 topics. This is done based on reasonable prior expectations about the dimensionality of the domain that we cover. This removes an additional 21 BERTopic fits and leaves us with 40 BERTopic fits.

Next, we sort the remaining 40 BERTopic fits by the DBCV metric (Moulavi et al. 2014). We then manually evaluate each of the top 10 candidates (highest DBCV) for the best separation of 4 topics that are expected and significant: "family eats meat", "protein (I want, need)", "animal welfare (concerned)", "environment (concerned)". Based on manual evaluation, we decide on the `qwen3-embedding-0.6b` model run with $neighbors = 30$, $components = 2$, and $cluster = 50$. This is the main model that we base the results reported in Section 4.2 on, but wherever reasonable, we report results for the 10 best BERTopic fits (based on DBCV) that satisfy our criteria. Key statistics for each model reported in 1 below, where `01__qwen3-embedding-0.6b` is the model that we select and primarily report.

Model	K	$\bar{\phi}_{\text{self}}$	$\bar{\phi}_{\text{other}}$	$\Delta\bar{\phi}$	AUC
`01__qwen3-embedding-0.6b`	27	0.36	0.10	0.26	0.80
`02__bert-base-nli-mean-tokens`	41	0.34	0.14	0.20	0.76
`03__bge-large-en-v1.5`	40	0.34	**0.09**	0.25	0.81
`04__bge-large-en-v1.5`	22	0.39	0.11	**0.28**	0.80
`05__all-MiniLM-L6-v2`	24	0.42	0.14	0.28	0.81
`06__bert-base-nli-mean-tokens`	23	0.38	0.16	0.21	0.75
`07__all-MiniLM-L6-v2`	25	0.39	0.11	0.28	**0.82**
`08__bge-large-en-v1.5`	24	0.38	0.11	0.28	0.80
`09__bert-base-nli-mean-tokens`	15	**0.43**	0.20	0.23	0.72
`10__bge-large-en-v1.5`	16	0.39	0.12	0.27	0.77

Table 1: Table of the 10 selected models (including outliers) showing the resulting number of topics K, the average phi coefficient within-individuals (across waves) $\bar{\phi}_{self}$, the average phi coefficient across-individuals (across waves) $\bar{\phi}_{other}$, $\Delta\bar{\phi} = \bar{\phi}_{self} - \bar{\phi}_{other}$, and the AUC (or probability of superiority). Higher AUC and $\Delta\bar{\phi}$ indicate better separation of within-individual relative to across-individual semantic content. Models are sorted by DBCV (Moulavi et al. 2014). No model is better across all metrics, but all models show a robust separation between within-individual and across-individual statistics.

We now have a mapping between all 3458 stances (extracted from 2×210 interviews) and 27 topics (including the outlier topic). The mapping between stances and topics is really a probabilistic assignment (a stance is a distribution over topics), but here we use a binary mapping to the most dominant topic for each stance. We also include the outlier topic because removing it can break the structure of the individual belief networks that we measure. Both choices are conservative as they force us to not filter out noise introduced in the mapping process.

Below, we present an overview table of the $n = 27$ topics (including outlier) for the selected model (`01__qwen3-embedding-0.6b`). We generally observe reasonable separation of the target topics: (i) family eats meat (topics 7, 12, and 22), (ii) wanting protein (topics 3, 13), (iii) animal welfare concern (topic 4) and (iiii) environmental concern (topic 20). We observe two main issues across most topic models. The first is that some topics that seemingly should be contained in one topic (e.g., protein from meat) end up in multiple topics. The other is that the topic model does not always respect the implication for meat-eating that we are focusing on here. For instance, in the selected model, we have "I became vegetarian", and "No vegetarians in my social circle" in the same topic (topic 3). This is not unreasonable from a "topic" point of view, but highlights that topic models do not necessarily capture the "direction" (or stance).

Topic	n	Keywords	Representative Docs
-1	664	OUTLIER	I enjoy eating meat, Meat-eating is a family tradition, I have no moral issues eating meat
0	244	chicken, fish, eat chicken, turkey, eat, chicken fish, chicken eat, chicken turkey, prefer, eat fish	I mostly eat chicken, I eat chicken, beef, and pork, I alternate beef and turkey weekly
1	232	consumption, meat, red meat, red, meat consumption, limit, try, cut, reduce, reducing	Doctor recommended I eat lean beef, I have meatless meals to save money, I try to limit meat to once daily
2	210	expensive, meat, meat expensive, choices, prices, meat prices, cost, influences, convenient, eating	Red meat is expensive, Vegetarian options are unaffordable, Meat can be expensive for me
3	154	protein, meat protein, believe, protein meat, believe meat, meat, source, iron, eat meat protein, meat good	I eat meat for protein, Meat is high in protein, Easier to get protein from meat
4	123	animal, welfare, animal welfare, concerned animal, concerned animal welfare, concerned, welfare concerned, animal welfare concerned, animals, welfare concerned animal	I am concerned about animal rights, Concern for animal welfare, I am concerned about animal welfare
5	122	husband, wife, meat health, reasons, red meat, red, meat, meat husband, friends, partner	My wife prefers red meat, My mother is health conscious, My husband eats less red meat
6	111	want, weight, motivated, weight want, lose, motivated health, lose weight, healthy, health, lose weight want	I am motivated by health concerns, I want to stay healthy, I want to be healthy
7	110	family, family eats, family eats meat, eats, eats meat, meat family, meat family eats, meat, eats meat family, eats meat regularly	My family eats meat, My family eats meat like I do, Family plans meals around meat
8	110	vegetarian, vegan, vegetarian meals, vegetarians, meals, years, vegan vegetarian, friends, vegetarian vegan, veganism	I enjoy veggie burgers, I became vegetarian, No vegetarians in my social circle
9	107	health, cholesterol, concerned health, concerned, concerns, health concerns, health concerned, heart, concerned health concerned, cholesterol concerned	I have health concerns, My health focus has increased with age, I discuss health risks with my friend

Continued on next page

Topic	n	Keywords	Representative Docs
10	105	avoid, avoid eating, pork, pork avoid, meats avoid, processed, avoid pork, avoid red meat, avoid red, meat avoid	I avoid too much processed meat, I avoid big portions of meat, I avoid factory-farmed meat
11	99	health, concerns, red meat, red, unhealthy, health concerns, meat, meat unhealthy, digestive, concerns red	I believe pork is unhealthy, Health risks are associated with eating too much meat, I think excess meat is unhealthy
12	96	friends, family, friends eat, friends eat meat, meat friends, family friends, friends family, family friends eat, friends family eat, eat meat	My parents served meat with dinner, My family and friends eat a lot of meat, Family and friends eat much more meat
13	94	protein, want protein, want, protein want, protein want protein, want protein want, need, need protein, protein diet, high protein	I want to get enough protein, I want to hit protein goals, I want to consume a lot of protein
14	93	times, week eat, week, meat times, eat meat times, week eat meat, weekly, eat, weekly eat, times week	I eat beef four or five times weekly, I eat red meat up to twice a week, I eat meat once or twice a week
15	92	dislike, meals, picky, picky eater, meat, like, meatfree, eater, meat dislike, meatless	I like exploring meatless dishes, I do not eat meat, I have never considered not eating meat
16	88	environmental, concerned, impact, environmental impact, concerned environmental, environment, concerned environmental impact, impact concerned, environmental impact concerned, environment concerned	I care about sustainability, I care about the environment, I am concerned about environmental impacts
17	84	like taste meat, like taste, meat meat, like, taste meat meat, taste meat, meat tastes, taste, tastes, incomplete	Meat tastes great with meals, Meat tastes good to me, Meat goes well with foods like pasta or rice
18	76	day eat, day, day eat meat, meat day eat, meat day, eat meat day, daily eat, daily, daily eat meat, eat meat	I eat meat every day, I eat one large meal daily, I eat bacon every day
19	74	meat meals, meals eat meat, meat meal, meals eat, meals, eat meat meals, meat meals eat, eat meat, meal, eat meat meal	I eat meat for dinner most days, I usually have meat at breakfast, I eat meat with most meals
20	72	plantbased, vegetables, plantbased meals, balance, balance meat, meals, eat plantbased, eat vegetables, plantbased foods, try	I want lighter or more vegetable meals sometimes, I enjoy vegetables and salads, I include plant-based meals for health
21	67	crave, meat crave, like, crave meat, meat, different, kinds meat, kinds, meat eat, available	I eat meat when craving it, I crave meat sometimes, I really like processed meats
22	63	grew, grew eating, raised, grew eating meat, raised eating, eating, eating meat, meat grew, raised eating meat, meat grew eating	I was raised eating pork, I grew up in a hunting family, I grew up eating meat-centered meals
23	60	eat meat regularly, meat regularly, regularly, meat regularly eat, regularly eat, regularly eat meat, eat meat, meat eat meat, eat, meat eat	I eat various meats regularly, I eat meat regularly, I have always been a meat eater

Continued on next page

Topic	n	Keywords	Representative Docs
24	56	diet, balanced, balanced diet, want, variety, eat healthier, mindful, healthier, want balanced, food	Doctor required me to watch fat intake, I cannot stick to eating only salads, I want proper nutrition
25	52	enjoy taste meat, enjoy taste, meat enjoy, taste meat enjoy, meat enjoy taste, enjoy, taste meat, taste, meat enjoy flavor, flavor	I enjoy the taste of meat, I enjoy the flavor of some meats, I enjoy pleasant meat aromas during cooking

7.4 LLM Prompts

7.4.1 LLM semi-structured interview prompt

The prompt for the semi-structured LLM interview is shown below. This prompt is used iteratively to generate a new question based on all previous questions and answers.

Prompt: generate_conversational_question

```
def generate_conversational_question(history: List[UserAnswer],
    n_rounds=8) -> InterviewTurn:
    conversation_str = ""
    for turn in history:
        conversation_str += f"Interviewer: {turn.question}\nParticipant:
    {turn.answer}\n\n"

    current_round = len(history) + 1

    system_prompt = f"""
Context:
You are a thoughtful, empathetic, and curious interviewer exploring the
    meat-eating habits and motivations of an interviewee.

Current conversation:
{conversation_str}

=*=*=

Task Description:

Interview objective: By the end of the conversation, the interviewer has
    to learn about the following:
1) The participant's personal meat-eating habits.
2) The participant's personal motivations for eating meat and/or for
    reducing/avoiding it.
3) The meat-eating habits of the interviewee's social contacts (e.g.,
    family, friends, peers, community).
4) The motivations that the interviewee's social contacts have to eat or
    avoid meat.

Follow this strategy to generate your next question:
```

```
1) Assess which of the 4 coverage goals have been adequately addressed so
    far, and which ones still need more exploration.
2.1) If the previous answer introduced something potentially (1)
    important, (2) interesting, or (3) unclear, formulate an elaboration
    question about this.
2.2) Otherwise, prefer asking about an uncovered goal from the list above.
2.3) If one the the interview goals above is not important for the
    participant do not pursue it further.
3) For the last question, invite the participant to share anything
    important that has not yet come up.

Follow these guidelines when constructing your next question:
1) Acknowledge the participant's last answer to show you are listening and
    value their input.
2) Respond naturally to what the participant has said: be curious, warm
    and non-judgmental.
3) Ask one focused open question per turn (avoid asking about more than
    one thing and avoid leading phrasing).
4) Keep it concise: ~1 sentence acknowledging what they said, then 1 clear
    question.
5) Avoid moralizing, advice, assumptions, checklists, multiple-choice, or
    multi-part questions.

Safety note: In an extreme case where the interviewee *exlicitly* refuses
    to answer the question do not force the interviewee to answer. Instead
    move the interview forward by asking about another topic from the list
    above.

Conversation constraints:
- You have {n_rounds} total turns; this is round {current_round} of
    {n_rounds}.

Based on the current conversation generate the next interviewer question
    that best follows the strategy and guidelines above.
    """

    return call_openai(
        response_model=InterviewTurn,
        content_prompt=system_prompt,
        model_name='gpt-4.1-2025-04-14',
        temp=0.7
    )
```

7.4.2 LLM summary statement prompt

The prompt for extracting summary statements (nodes) based on the interview transcript is given below. This is executed between the semi-structured interview and the evaluation of the *accuracy* and *relevance* of the extracted summary statements.

Prompt: make_node_prompt

```
def make_node_prompt(questions_answers: Dict[str, str]) -> str:
    questions_answers_lines = "\n".join(
        f"- Q: {q}\n- A: {a}" for q, a in questions_answers.items()
    )

    prompt = f"""
Context:
You are an expert analyst specializing in extracting key factors
   influencing meat consumption from interview transcripts.

Interview Transcript:
{questions_answers_lines}

=*=*=

Task description:
Analyze the interview transcript and extract **only** factors that
   influence the meat-eating habits of the interviewee.
Extract both (1) factors that encourage or increase meat-eating, and (2)
   factors that discourage or reduce meat-eating.

For each factor, determine whether it relates to:
1) the personal behaviors and motivations of the interviewee, or
2) the behaviors and motivations of their social contacts (e.g., family,
   friends)

Only extract factors that relate to the behaviors and motivations of
   social contacts when these plausibly influence the meat-eating habits
   of the interviewee.

=*=*=

Extraction rules:
1. Extract a maximum of 10 short statements (max 10 words each). If there
   are fewer than 10 relevant factors, extract only those that are clearly
   relevant.
2. Extract at least 1 statement about the personal meat-eating habits of
   the interviewee.
3. Provide the following fields for each statement:
- **stance**: A concise summary of the attitude or behavior (no effects).
- **type**: Is the statement about the interviewee <PERSONAL> or about
   social contacts <SOCIAL>.
- **category**: Does the statement describe a <BEHAVIOR> or a <MOTIVATION>.
- **effect**: Does the statement describe something that <INCREASE> or a
   <DECREASE> the meat-eating habits of the interviewee.
4. If you cannot reliably assign an effect (increase or decrease), do not
   include the statement.

=*=*=
```

```
Extraction guidelines:
1. Include only relevant influences:
    - Exclude anything that is unimportant, irrelevant, or does not
   clearly have an effect.
    - Example (EXCLUDE): "Health is not important to me."

2. Differentiate behavior from motivation:
    - Each statement should describe either a behavior or a motivation,
   not both.
    - **Behavior** = descriptions of behavior (e.g., "I never cook meat at
   home", "I eat beef every week").
    - **Motivation** = any reason to eat more or less meat (e.g., "I love
   the taste of meat", "I am concerned about animal welfare").
        - Example: "I avoid meat because of health concerns" should be
   split into two separate statements ("I avoid meat" and "I have health
   concerns").
        - Example: "Living with vegetarians limits meat cooking at home"
   should be split into two separate statements ("I live with vegetarians"
   and "I rarely cook meat at home").
        - Example: "I eat beef to get enough protein" should be split into
   two separate statements ("I eat beef" and "I want to get enough
   protein").
        - Example: "I was vegetarian but now eat meat for weight gain"
   should be split into two separate statements ("I was vegetarian" and "I
   eat meat" and "I want to gain weight").
        - Avoid [due to, because, as a result of, since, for, etc.] ---
   split into separate statements **very important**.

3. Ensure that each statement is well formed in isolation and can be
   evaluated by the interviewee as something they agree with or not:
    - Each statement should be a clear, concise, and factual statement
   (max 10 words).
    - Each statement should be something that the interviewee plausibly
   agrees with.
    - Think "I agree with the following: [statement]" where you take on
   the perspective of the interviewee.
    - Not good: "Financial limitations make it difficult" [unclear, cannot
   agree or disagree] --> good: "I have financial limitations"
    - Not good: "Availability of good vegetarian options" [unclear, cannot
   agree or disagree] --> good: "Good vegetarian options are limited"

=*=*=

Example:

Input transcript excerpt:
Q: Can you tell me what influences how much meat you eat?
A: I love the taste of meat, but I live with vegetarians who are very
   concerned about animal welfare, so I rarely cook it. My family always
```

```
    serves meat when I visit. I also worry about the climate impact of
    eating too much meat.

Expected output:
{{
"results": [
    {{
        "stance": "I love the taste of meat",
        "type": "PERSONAL",
        "category": "MOTIVATION",
        "effect": "INCREASE"
    }},
    {{
        "stance": "I live with vegetarians",
        "type": "SOCIAL",
        "category": "BEHAVIOR",
        "effect": "DECREASE"
    }},
    {{
        "stance": "Social contacts are concerned about animal welfare",
        "type": "SOCIAL",
        "category": "MOTIVATION",
        "effect": "DECREASE"
    }},
    {{
        "stance": "I rarely cook meat",
        "type": "PERSONAL",
        "category": "BEHAVIOR",
        "effect": "DECREASE"
    }},
    {{
        "stance": "My family serves meat during visits",
        "type": "SOCIAL",
        "category": "BEHAVIOR",
        "effect": "INCREASE"
    }},
    {{
        "stance": "I am concerned about climate change",
        "type": "PERSONAL",
        "category": "MOTIVATION",
        "effect": "DECREASE"
    }}
]
}}

=*=*=

### Output Format (JSON ONLY) ###
{{
"results": [
```

```
    {{
        "stance": "<concise summary of one motivation or behavior>",
        "type": "<PERSONAL or SOCIAL>",
        "category": "<BEHAVIOR or MOTIVATION>",
        "effect": "<INCREASE or DECREASE>"
    }}
]
}}
Return ONLY the JSON object, nothing else.
"""
    return prompt
```

7.4.3 LLM connections prompts

The prompt for extracting connections between participant-accepted summary statements based on the interview transcript a list of these nodes. This is executed during the survey and is used for the network comparison (see Section 2.5 and SI Section 7.1.5). The prompt is given below.

Prompt: make_edge_prompt

```
def make_edge_prompt(
    questions_answers: dict,
    nodes_list: list,
):
    # Format Q&A block
    questions_answers_lines = "\n".join(
        f"Q: {q}\nA: {a}" for q, a in questions_answers.items()
    )

    # Format the nodes
    belief_string = "\n".join(f"- {x}" for x in nodes_list)

    prompt = f"""
# Context:
You are an expert analyst specializing in understanding the relationships
   between beliefs, motivations, and behaviors related to meat consumption.
You are given a transcript of an interview along with a list of extracted
   stances (behaviors, beliefs, motivations, reasons, concerns, etc.).

# Interview transcript:
{questions_answers_lines}

=*=*=

# Extracted stances:
{belief_string}

=*=*=

# Task description:
```

```
Analyze the interview transcript and the list of extracted stances to
    identify meaningful connections between the stances.
You are looking for two types of relations:
1. Supportive connections, where one stance reinforces or encourages
    another.
2. Conflicting connections, where one stance contradicts or undermines
    another.
The goal is to build a network of how these stances relate to each other
    in terms of support or conflict.
The connections are undirected (i.e., A <-> B).

=*=*=

# Extraction rules:
1. Consider all unordered pairs of distinct stances (A, B).
2. If there is **strong evidence** for a supportive or conflicting
    relationship that is felt by the interviewee include the undirected
    edge (A, B).
3. For **every** included edge:
   - Use **alphabetical order** for the two stances to avoid duplicates.
   - Set **polarity** to positive (supportive) or negative (conflicting).
   - Set **explicitness** to explicit or implicit depending on whether is
    **directly stated** or only **implied** in the transcript.
   - Assign a **strength** (1-100) based on the evidence in the
    transcript. Be conservative and use lower values if evidence is thin or
    ambiguous. Drop edges with weak evidence (<20).
   - Provide a concise **justification** for how (A, B) either support or
    conflict.

=*=*=

# Example:

Input transcript excerpt:
Q: Can you tell me what influences how much meat you eat?
A: I love the taste of meat, but I live with vegetarians who are very
    concerned about animal welfare, so I rarely cook it. My family always
    serves meat when I visit. I also worry about the climate impact of
    eating too much meat.

Extracted stances:
- I love the taste of meat
- I live with vegetarians
- Social contacts are concerned about animal welfare
- I rarely cook meat
- My family serves meat during visits
- I am concerned about climate change

Expected output:
```

```
Expected output:
{{
"results": [
    {{
        "stance_1": "I live with vegetarians",
        "stance_1": "I love the taste of meat",
        "polarity": "negative",
        "explicitness": "explicit",
        "strength": 80,
        "justification": "Loving the taste of meat conflicts with living
    with vegetarians."
    }},
    {{
        "stance_1": "I live with vegetarians",
        "stance_1": "Social contacts are concerned about animal welfare",
        "polarity": "positive",
        "explicitness": "implicit",
        "strength": 70,
        "justification": "The vegetarian diet of the co-inhabitants of the
    interviewee is supported by their concern for animal welfare."
    }},
    {{
        "stance_1": "I rarely cook meat",
        "stance_1": "I live with vegetarians",
        "polarity": "positive",
        "explicitness": "explicit",
        "strength": 90,
        "justification": "It is explicitly stated that living with
    vegetarians supports the interviewee rarely cooking meat."
    }},
    {{
        "stance_1": "Social contacts are concerned about animal welfare",
        "stance_1": "I rarely cook meat",
        "polarity": "positive",
        "explicitness": "explicit",
        "strength": 70,
        "justification": "It is explicitly stated that living with
    vegetarians who care about animal welfare supports the interviewee
    rarely cooking meat."
    }},
    {{
        "stance_1": "I love the taste of meat",
        "stance_1": "I rarely cook meat",
        "polarity": "negative",
        "explicitness": "explicit",
        "strength": 90,
        "justification": "Loving the taste of meat conflicts with rarely
    cooking meat."
    }},
    {{
```

```
        "stance_1": "I love the taste of meat",
        "stance_1": "My family serves meat during visits",
        "polarity": "positive",
        "explicitness": "implicit",
        "strength": 30,
        "justification": "Loving the taste of meat is plausibly implicitly
    supported by family environment"
    }},
    {{
        "stance_1": "I am concerned about climate change",
        "stance_1": "I rarely cook meat",
        "polarity": "positive",
        "explicitness": "explicit",
        "strength": 80,
        "justification": "Concern about climate change supports rarely
    cooking meat."
    }}
    {{
        "stance_1": "I love the taste of meat",
        "stance_1": "I am concerned about climate change",
        "polarity": "negative",
        "explicitness": "implicit",
        "strength": 50,
        "justification": "Loving the taste of meat conflicts with concern
    about climate change."
    }}
]
}}

=*=*=

# Output Format (JSON ONLY)
{{
  "results": [
    {{
      "stance_1": "<first node in alphabetical order>",
      "stance_2": "<second node in alphabetical order>",
      "polarity": "<positive|negative>",
      "explicitness": "<explicit|implicit>",
      "strength": <integer 1-100>,
      "justification": "<brief explanation>",
    }}
    // Repeat for each undirected edge
  ]
}}

ONLY return the JSON object, no additional text.
"""
    return prompt
```